汽车电子与控制
Automotive Electronics and Control Systems (AECS)

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2020.9

第一章：汽车电子与控制系统绪论
An introduction to Automotive Electronics and Control Systems

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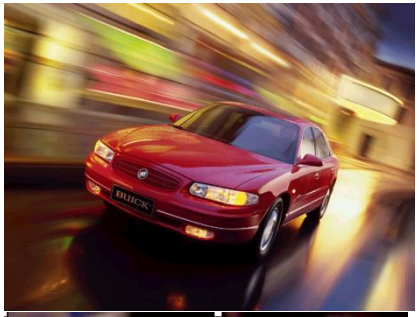
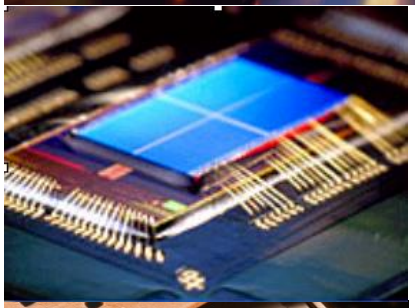
3汽车电子的基本框架
/Basic Framework of AE Control Systems

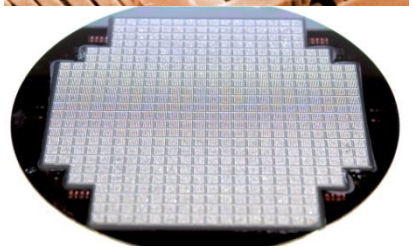
4汽车电子的开发流程
/Development Process of AECS

§1 汽车电子的技术特点
Basic Characteristics of Automotive Electronic Systems

特点/Feature:

- 适于高速移动/Suitable for High Speed: 200Km/h
- 满足苛刻和剧烈变化的环境要求/ Fulfill Critical Variation of Environment. (–40°~150°)
- 高度灵活性和高度可靠性/High Flexibility and High Reliability (10^{-8} /h Failure Rate)
- 机电耦合特性，软硬件结合/ Highly Integrated Mechanical and Electrical→ Mechatronic, High Integration of Software and Hardware.
- 适于大批量和低成本的生产/ Suitable for Mass Production and Low Cost



§1 汽车电子的技术特点
Basic Characteristics of Automotive Electronic Systems

汽车电子具有军品的质量和性能，家电的产量和价格/The performance of AES is similar to Military Electronics, meanwhile the cost of the AE is similar to Consumer Electronics.

汽车电子已经成为半导体行业一个独立的分支/As a result, the automotive electronics is glowing towards a special department for semiconductor industry.

从微电子工业到软件设计，从传感器和执行器的设计和制造，到测试和验证，汽车电子控制系统已经成为规模化生产的工业/From the Micro-electronic industry to the software design, from the design and manufacture of actuators and sensors, to test and validation, AEC has become large-scale industries independently.

§1 汽车电子的技术特点
Basic Characteristics of Automotive Electronic Systems

- 形成国际著名的大集团/Famous international Companies：博世/BOSCH,大陆/ Continental,德尔福/ Delphi, 电装/Denso, ……
- 飞思卡尔/Freescale, 英飞凌/Infineon, 意法半导体/ST, 微芯/Microchip, 瑞萨（Renesas）…….
- 如今，汽车电子的成本已经占到整车成本的35%~40%/ Nowadays, the cost of automotive electronics is about 35 % ~ 40 % of a vehicle.
- 汽车上90%的技术革新与汽车电子相关/ Almost 90% of innovation technique of Automotive is related to Automotive Electronics.

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3汽车电子的基本框架
/Basic Framework of AE Control Systems

4汽车电子的开发流程
/Development Process of AECS

（1）第一阶段：器件层次/Phase I：Simple Components.

分立式半导体元件开始用于汽车交流发电机整流器、固态收音机、电子点火、数字钟和车速控制电路等。/Discrete semiconductor components were used in the following parts of automobiles: rectifier of alternate generator, solid state radio, electronic ignition, digital clocks and control circuit of vehicle speed ,etc.

摩托罗拉：Motorola = Motor Radio

（2）第二阶段：部件层次/Phase II: System developed.

集成电路和4位、8位、16位微处理器(单片机)用来开发汽车专用的电子系统。/ICs and 4–bit、8–bit、16–bit micro–controllers and microprocessor were used to develop the automotive electronic systems.

包括发动机控制系统、自动变速箱控制系统、汽车防抱死制动系统/Including Engine Control System, Automatic Transmission Control System, Anti–lock Brake System et al.

（2）第二阶段：部件层次/Phase II: System developed.

以及电子悬架控制系统、音响控制系统、空调控制系统等/Electronic Suspension Control System、Audio Control System、Air Condition (Climate Control) System, etc.

主要的编程手段为汇编和C语言/ In this phase, the programming languages are assembly language and/or C/C++ language.

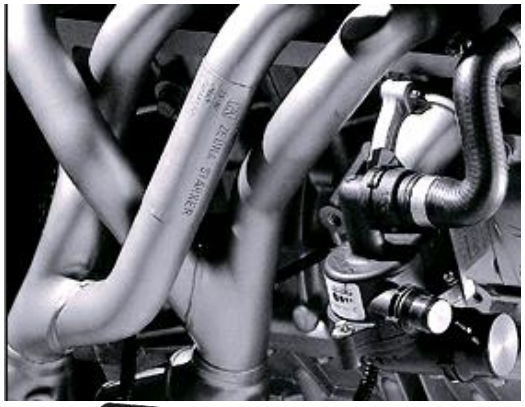
在这个阶段，工程师需要学习掌握汇编语言和C语言编程。

（3）第三阶段：整车集成/Phase III: Integration of Vehicle Control

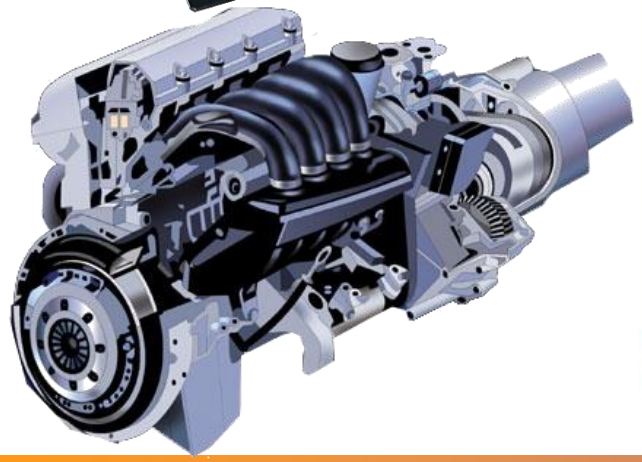
采用智能功率器件、智能传感器、大容量EEPROM或者FLASHROM、专用单片机等加入到汽车电子中/Smart Power Drive IC, Smart Sensors, EEPROM or FLASHROM with Large Capacity, Application Specific Integrated Circuit (ASIC) etc. are joined in the industry of automotive electronics.

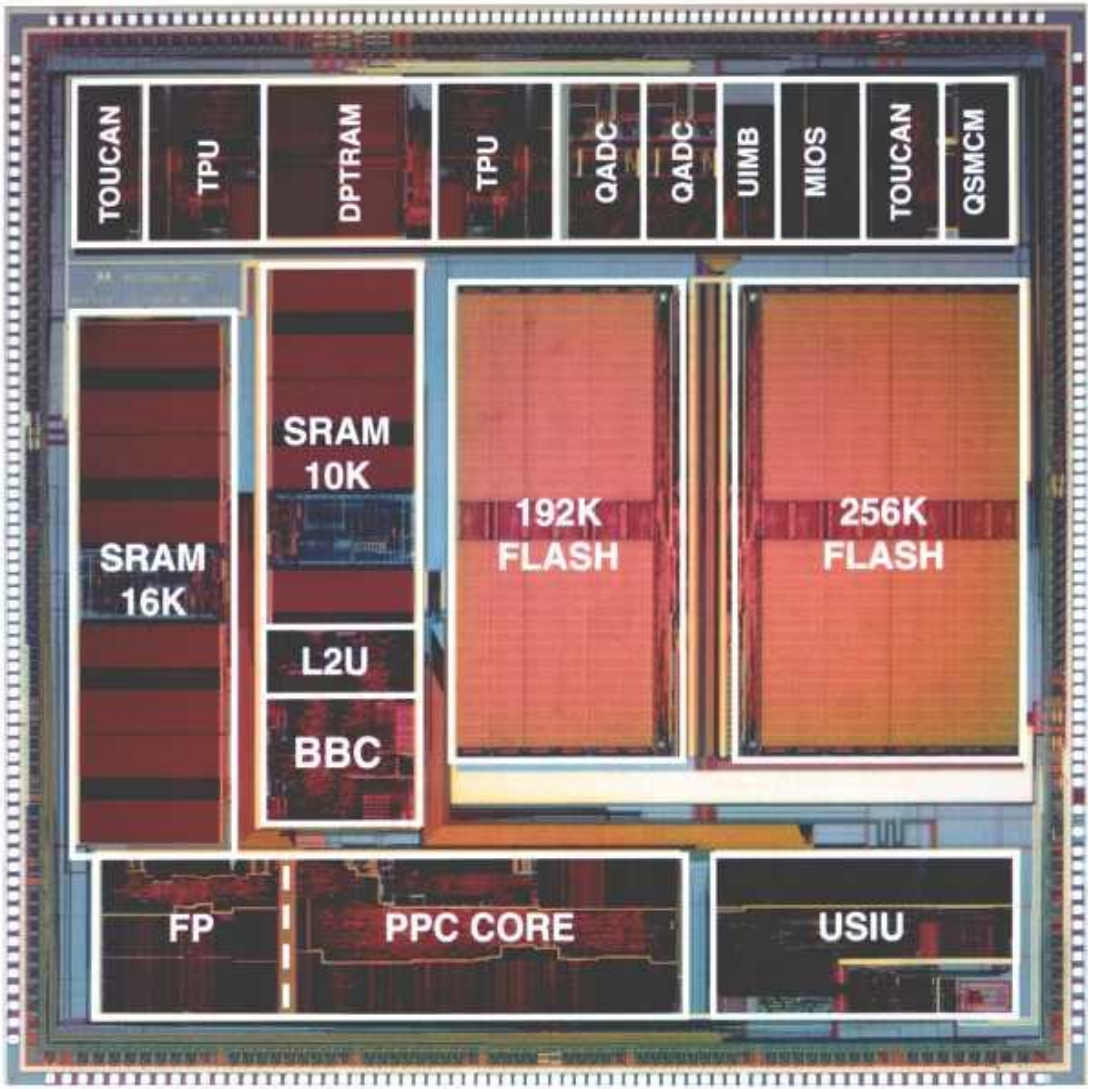
进一步形成了规范化、专业化的开发、生产和设计体系/ An architecture including development、production and design standardization and specialization has been built.

32位控制器/32bit MCU: PowerPC555

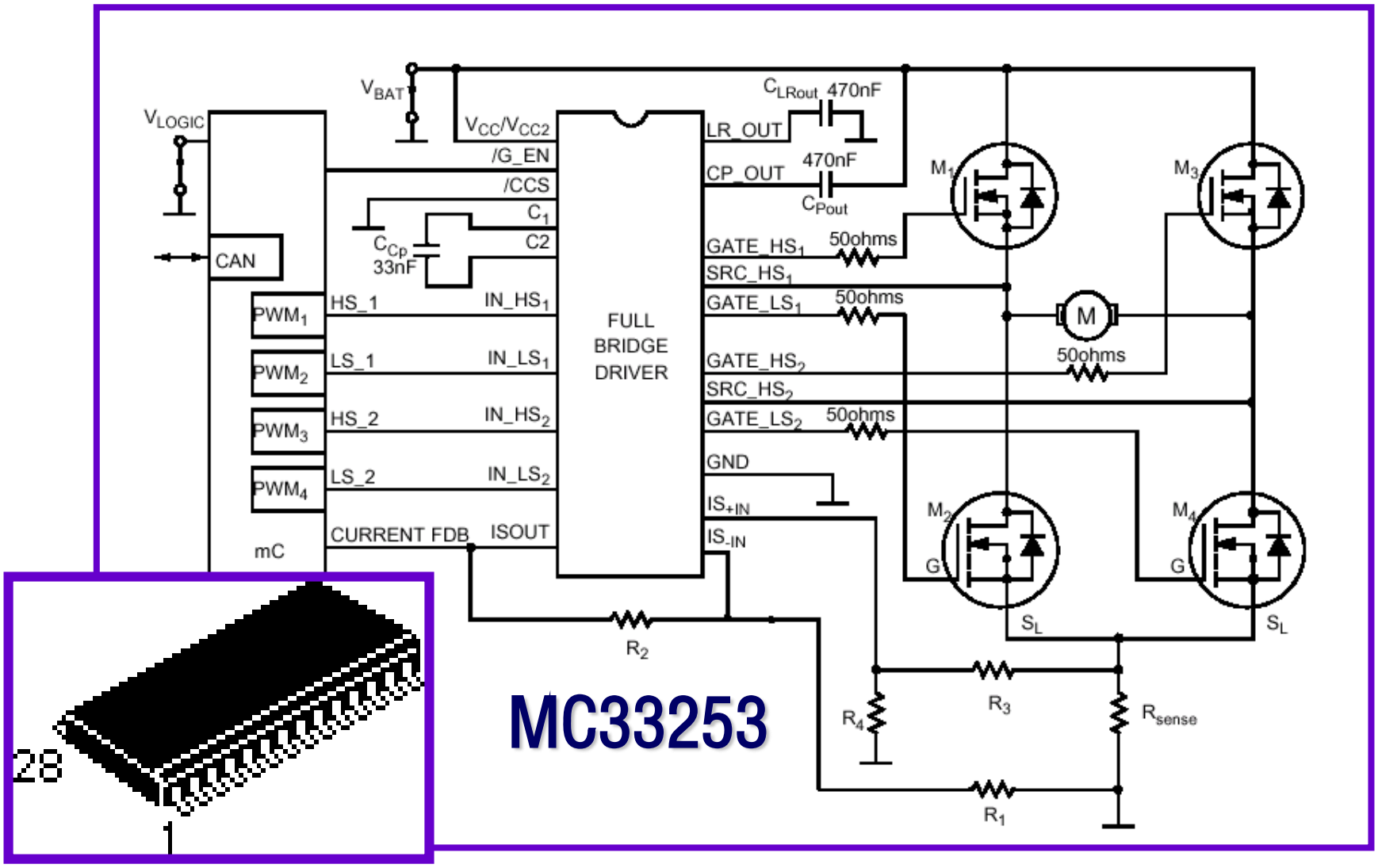








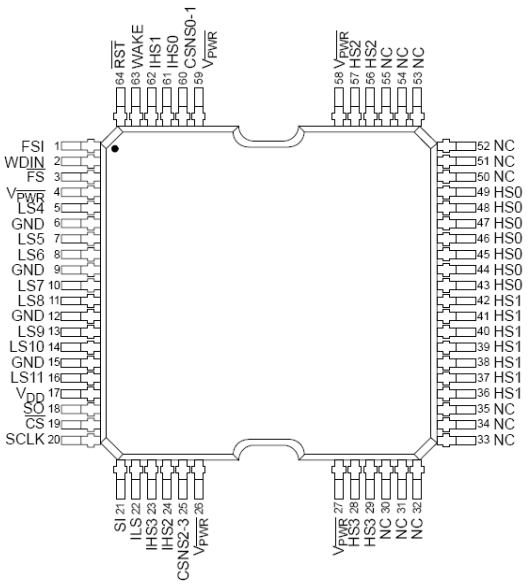
全桥预驱动器/Full Bridge Pre-Driver

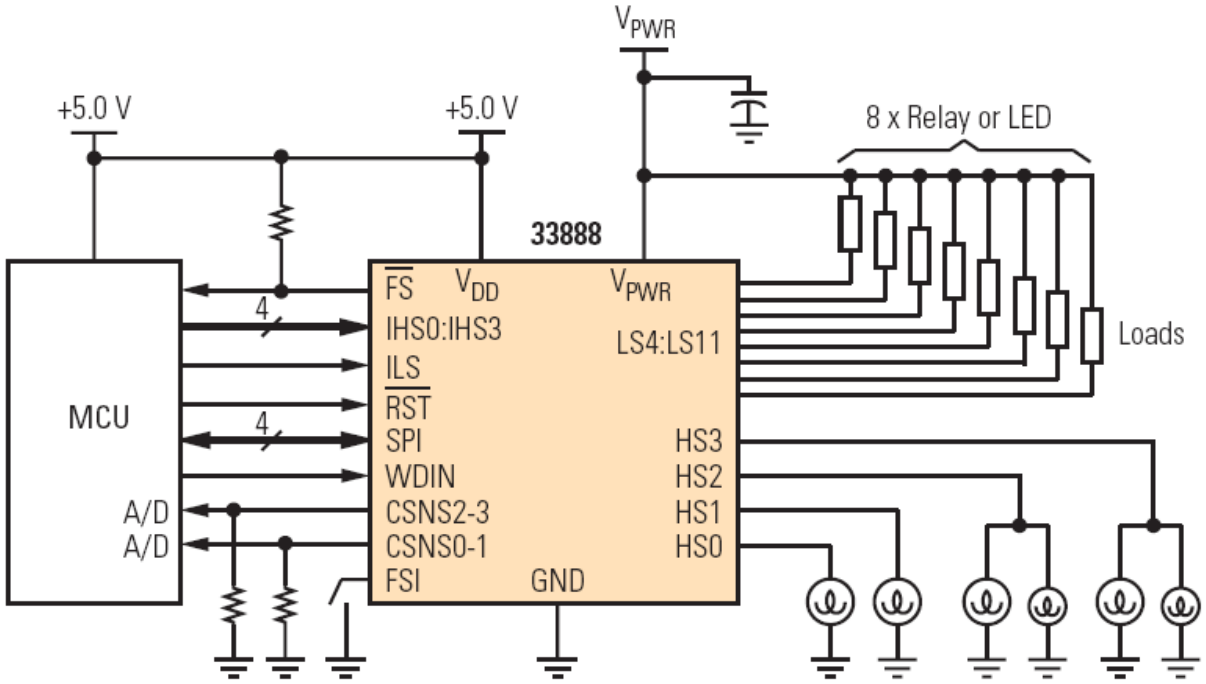


MC33888 高边/低边驱动/(High Side/Low Side Driver)

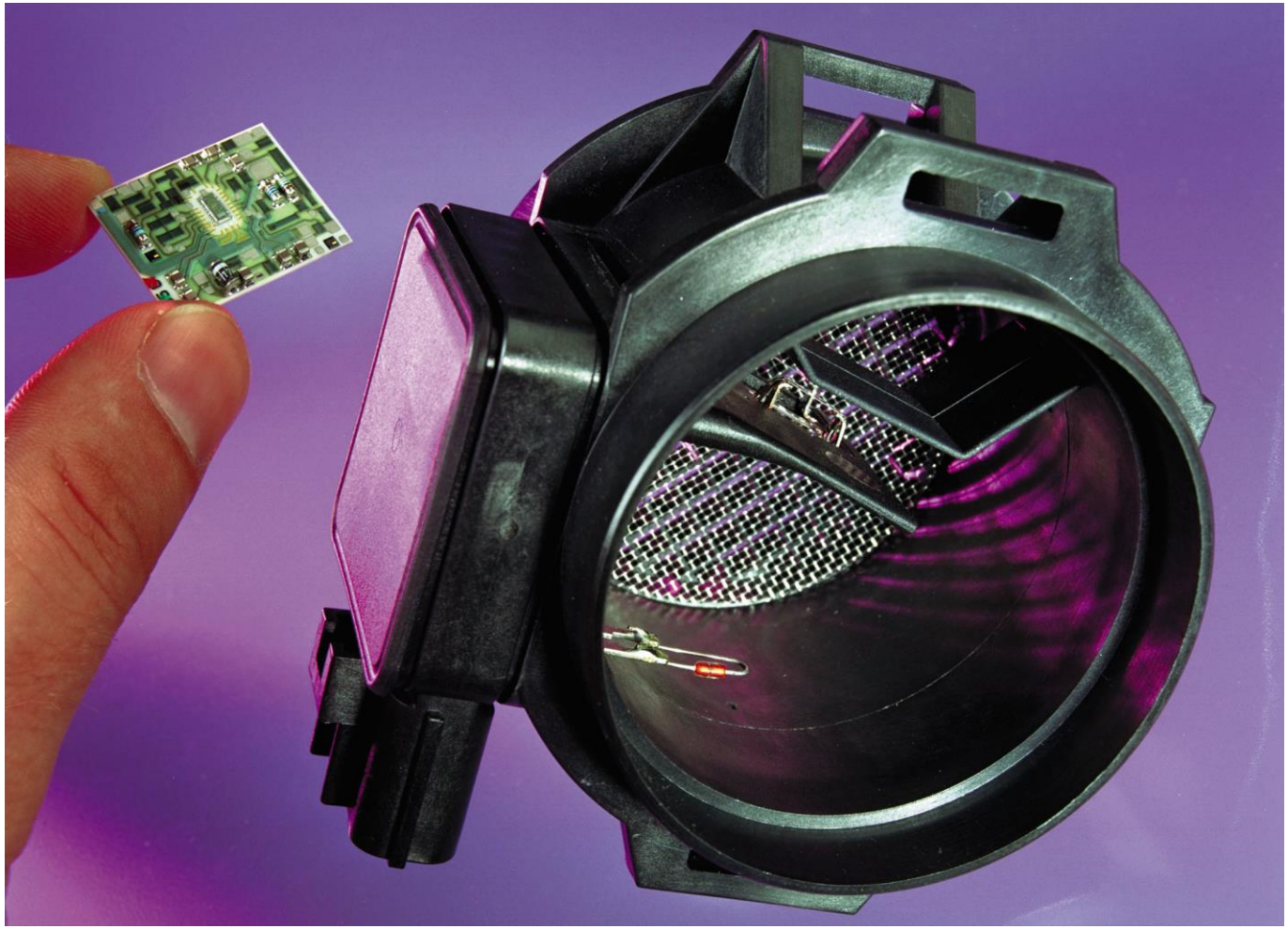


MO188
64 pins
0.65 pitch

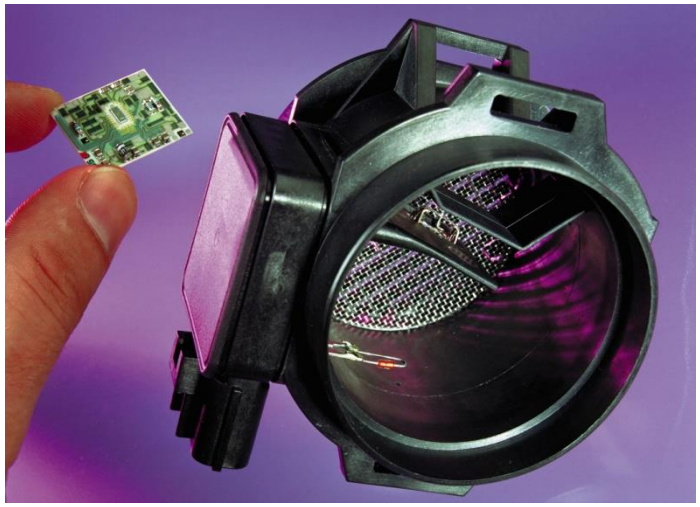


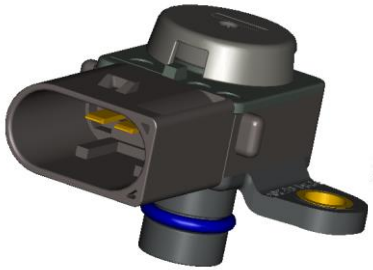


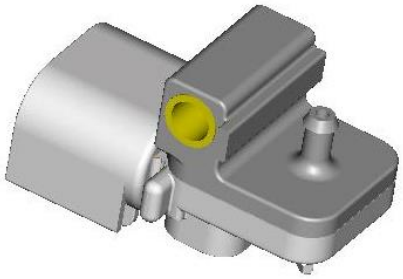
空气流量传感器/Air Mass Flow Sensor

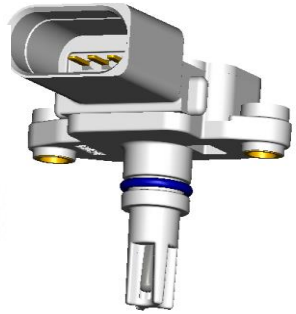


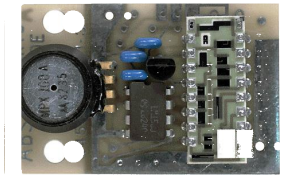
各种压力传感器/Variant Pressure Sensors

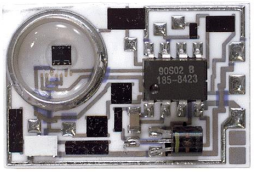


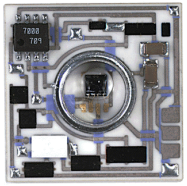


















1983

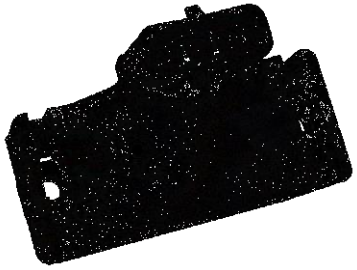
1985

1987

1991

1999

2005







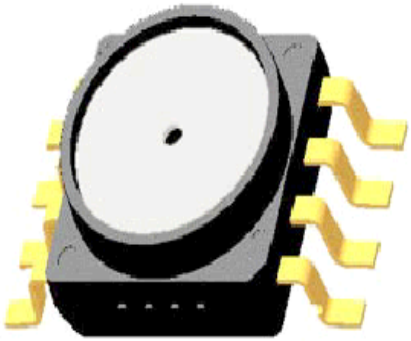


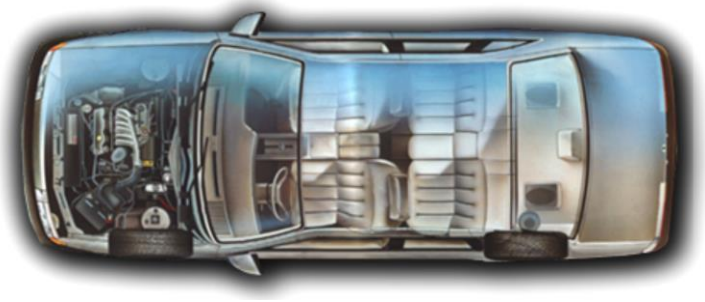


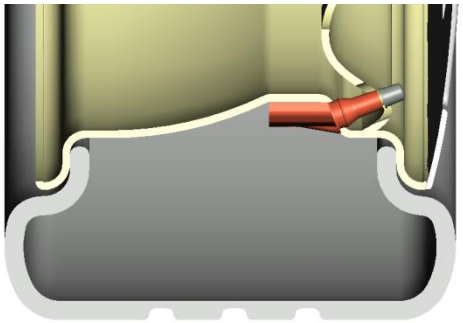


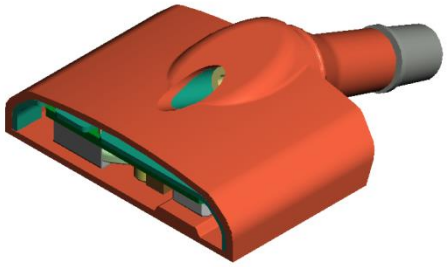
胎压监测系统/Tire Pressure Monitor System

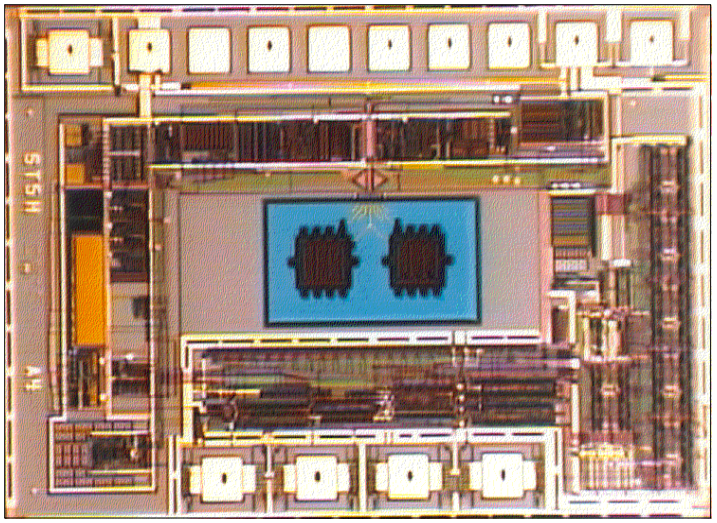


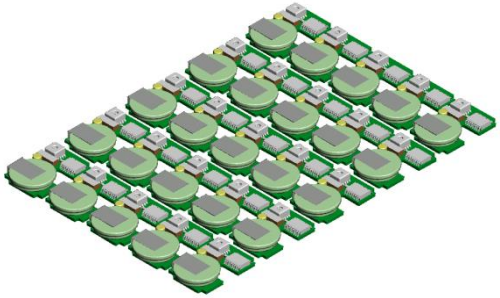




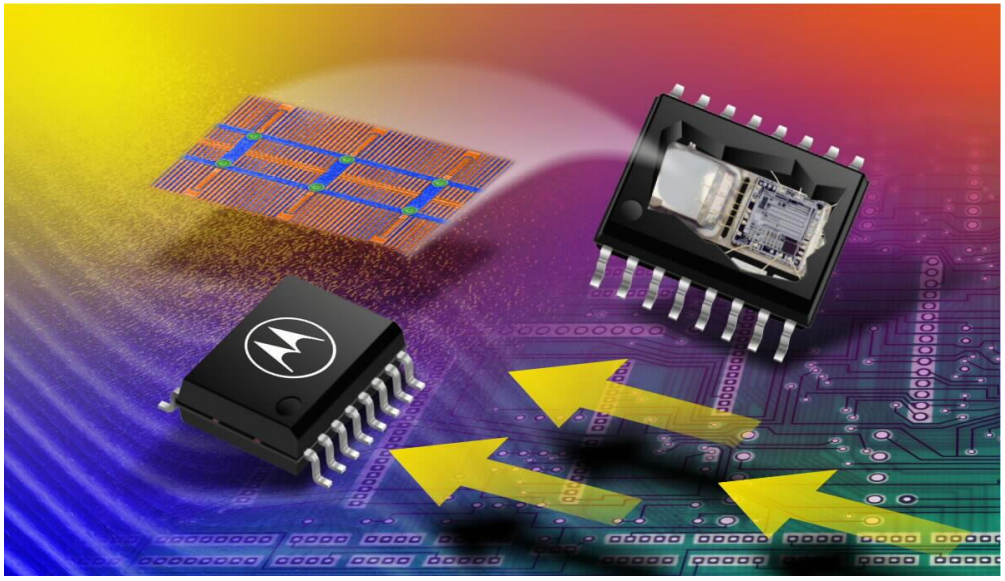


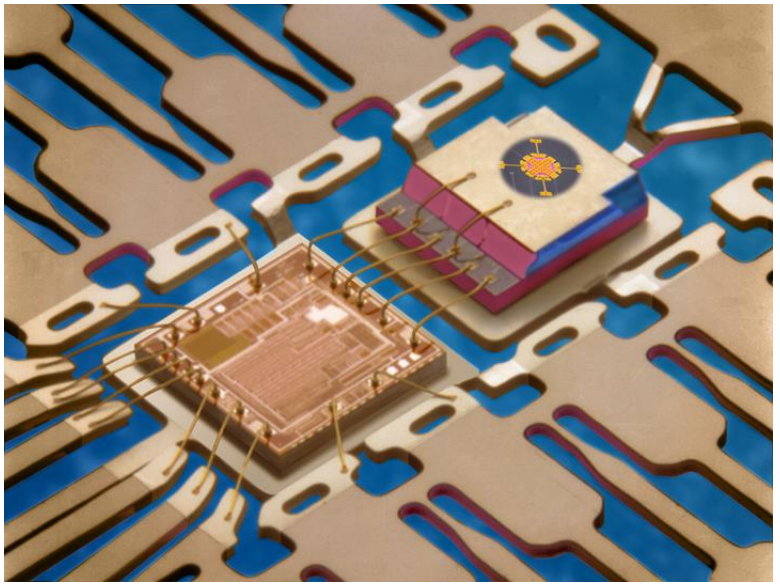


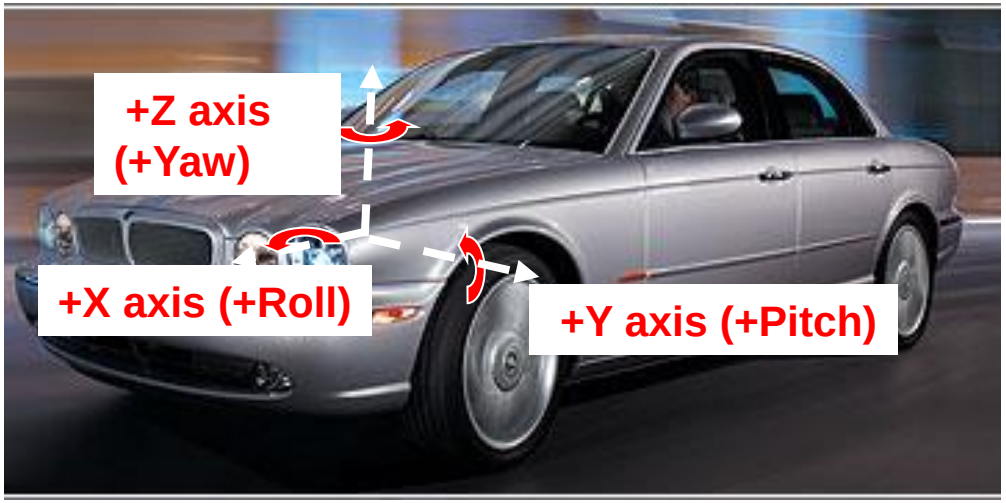


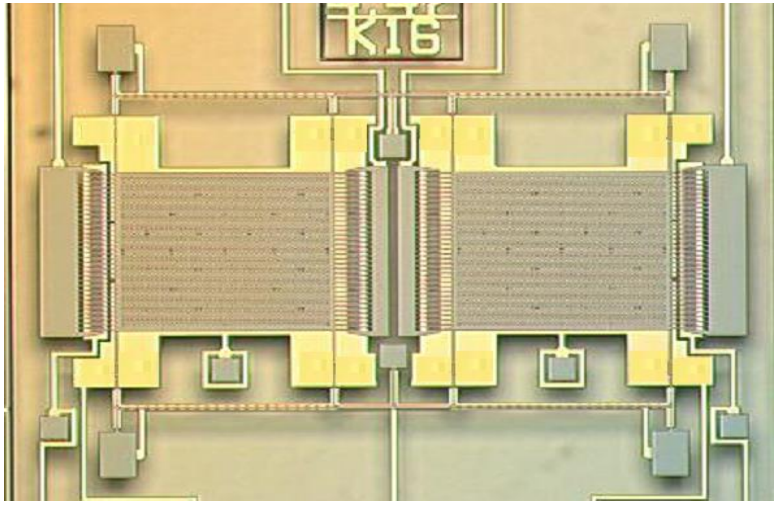


各种微机电传感器/Variant MEMS Sensor









（3）第三阶段：整车集成/Phase III

- 发动机和变速箱的动力总成综合控制（Powertrain Control, Drive by Wire）；
- 转向（Steering）和悬架（Suspension）综合控制系统(Steering by Wire)；
- 电子制动和防滑控制系统（ABS and ASR, Brake by Wire)
- 混合动力系统（Hybrid Powertrain System)
- 主被动安全控制系统（Active and Passive Safety Control System）；
- 智能巡航控制系统（Adaptive Cruise Control），
- 数字音响系统（Digital Radio）；
- 智能交通系统（Intelligent Transportation System）；
- 通讯导航和远程诊断（Guide and Navigation，Remote Diagnostic）；
- 直到整车综合控制系统（Integrated Vehicle Control System）；

X–by–Wire

形成以网络为基础的多ECU的分布式控制系统。（Network based Multi–ECU Distributed Control System）

分布式–域集中式–中央控制式

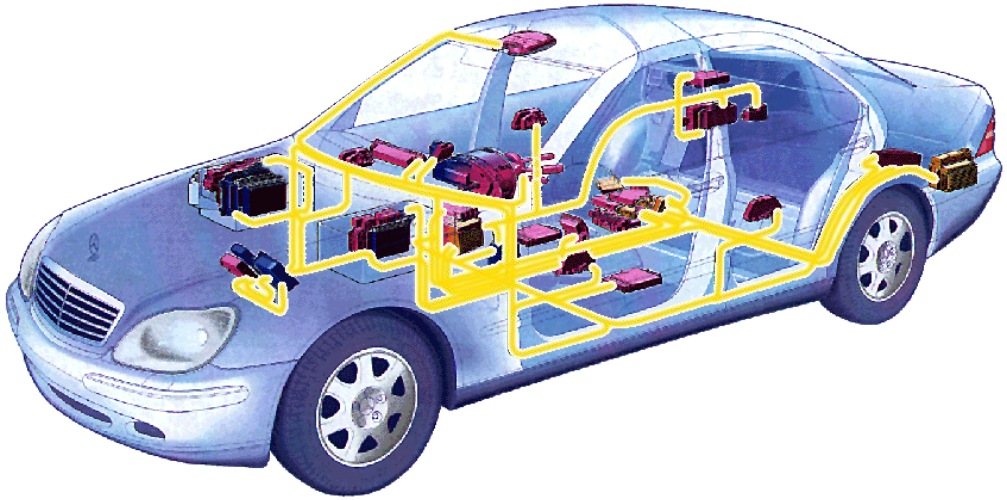
整车集成/Vehicle Integration



- Local Interconnect Network
→LIN (SCI)

- Controller
- Area
- Network
→ CAN

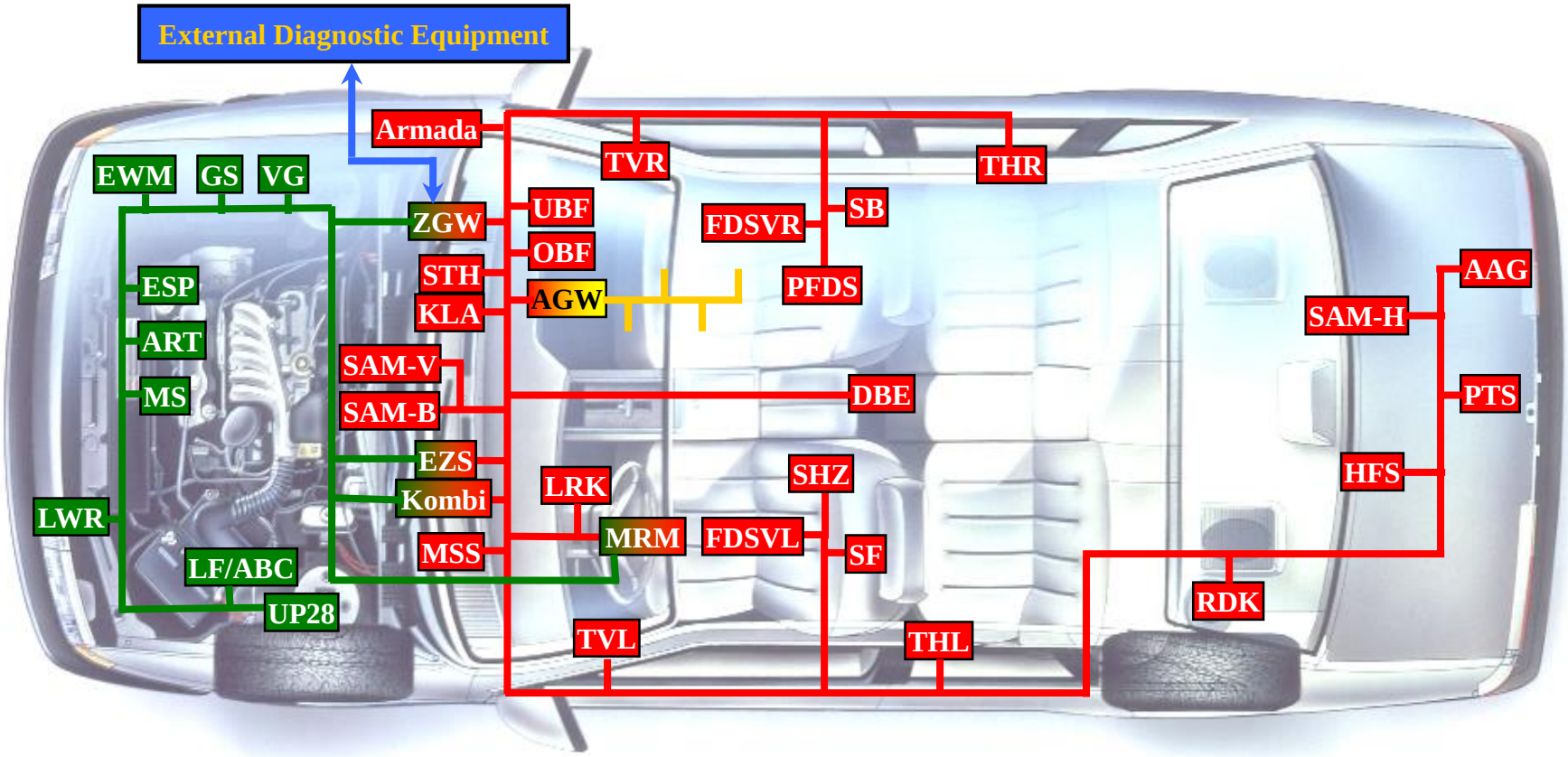
整车网络组成：分布式控制系统
Distributed Control System of Vehicle



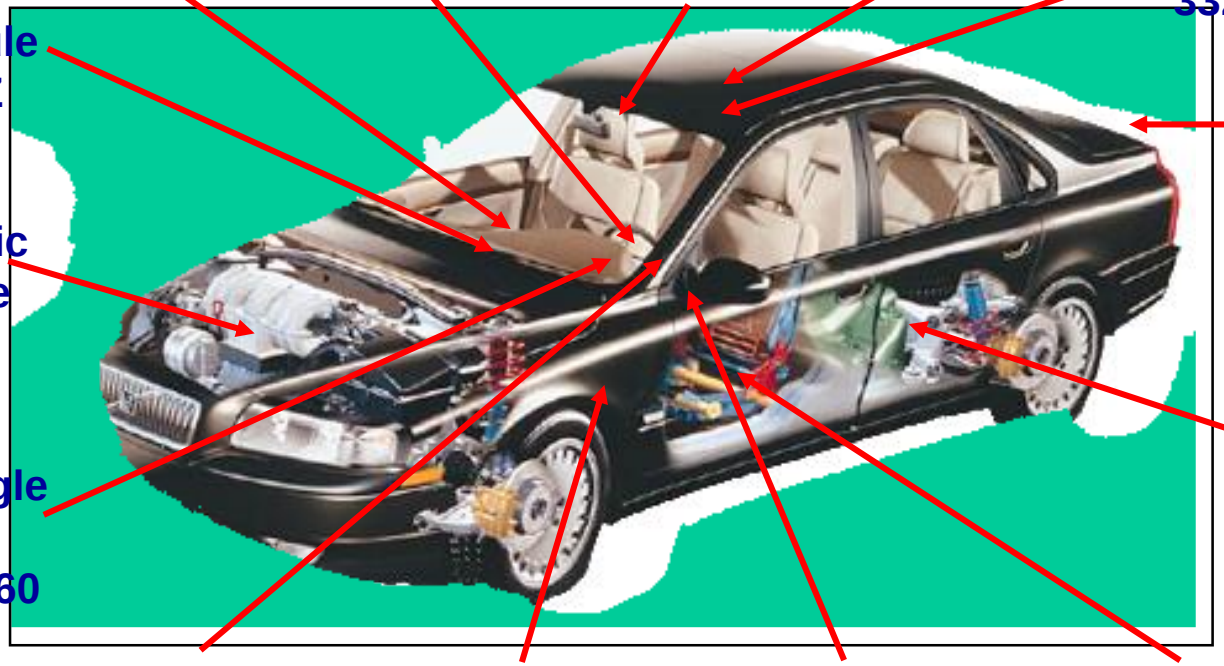


无线遥控与车载网络
Wireless key and Inner Vehicle Network

整车网络组成：分布式控制系统
Distributed Control System of Vehicle

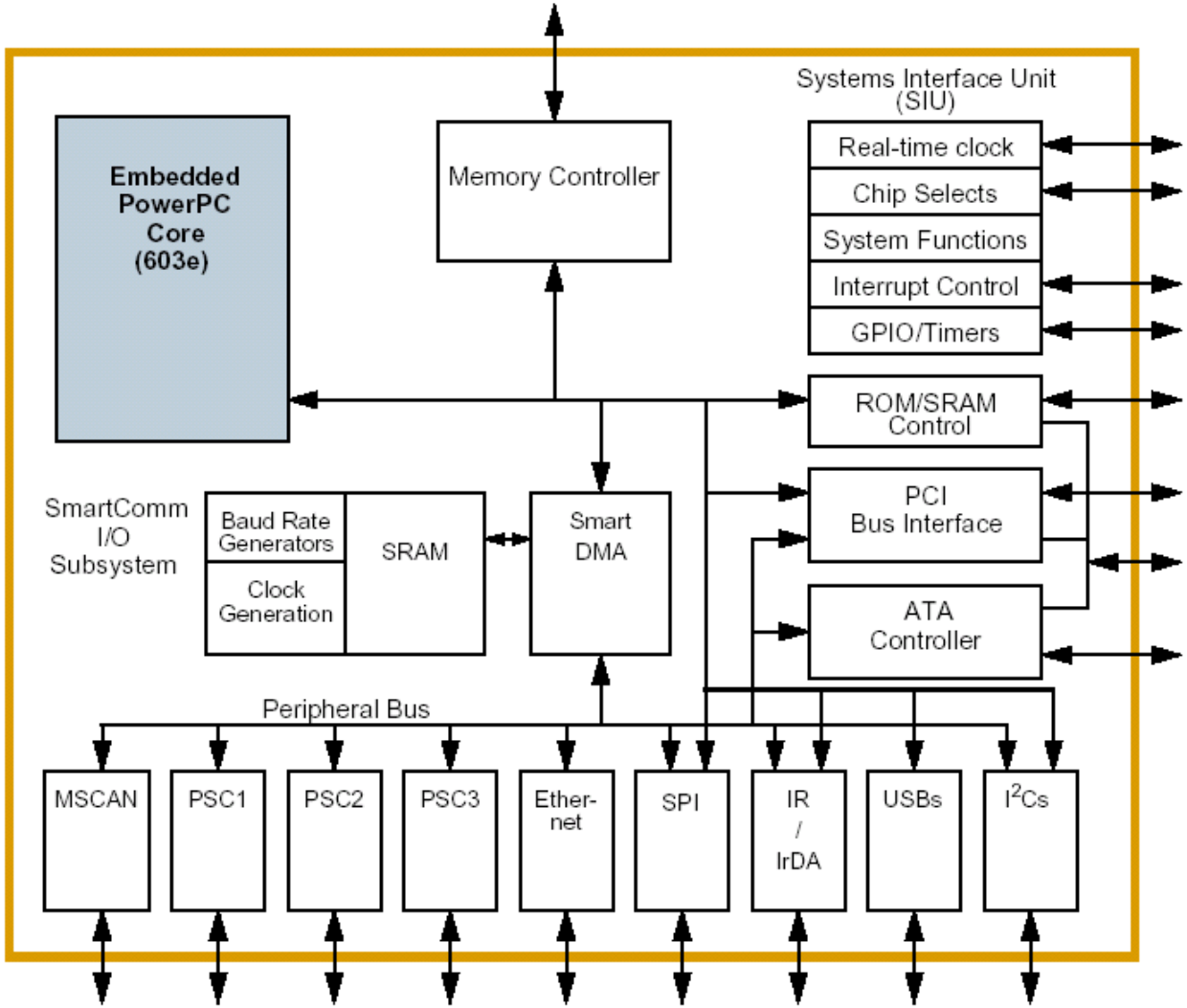


整车网络组成：分布式控制系统
Distributed Control System of Vehicle

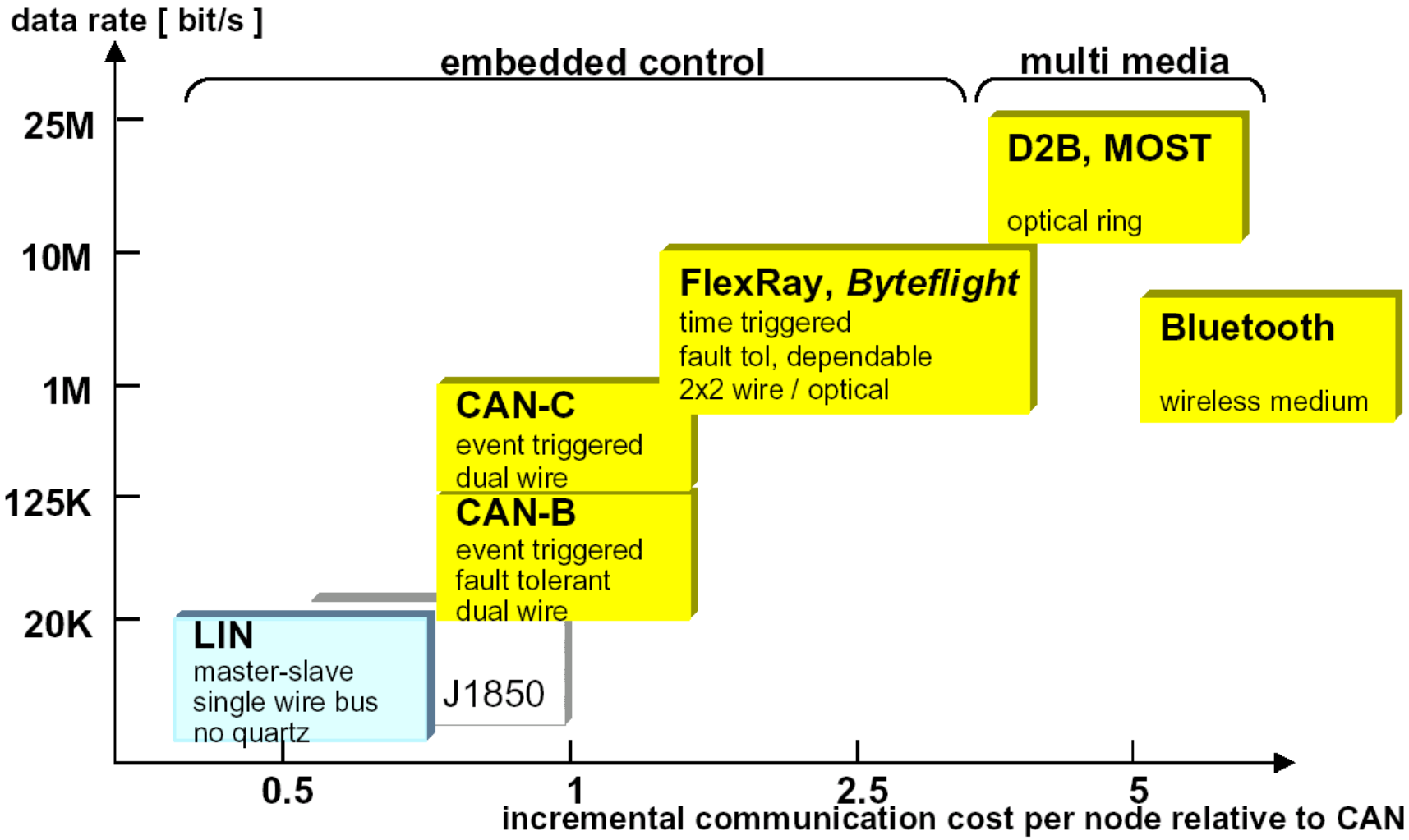


基于网络的汽车专用MCU
Network Based Automotive MCU

MGT5100



车载网络的发展趋势/
Network Trends In Automotive



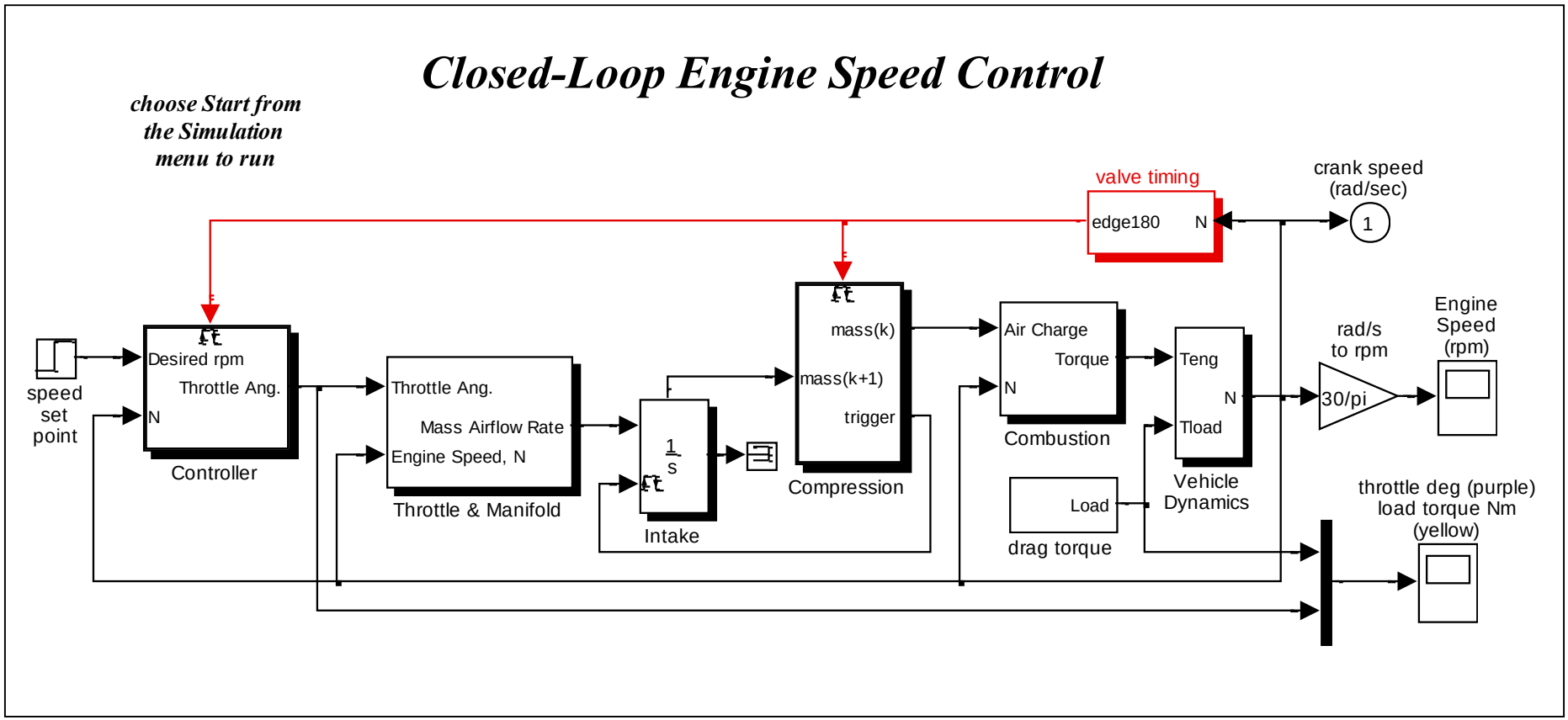
（3）第三阶段：整车集成/Phase III: Integration of Vehicle Control

●在软件的开发手段上，采用了MATLAB/ Simulink /Stateflow作为仿真和调试手段，dSPACE作为ECU的快速原型开发代表性技术，以图形化界面编程为主的手段，以嵌入式实时操作系统RTOS为框架，形成OSEK/VDX的统一标准和规范。/In the software design, MATLAB/ Simulink /Stateflow was used to simulate and validate. The technology of ECU rapid prototyping such as dSPACE, by programming with graphic interface, under the platform of embedded real time operation system (RTOS) , formed the stand alone executable code of the MCU and fulfills the standards of OSEK/VDX.

MATLAB/SIMULINK 仿真举例

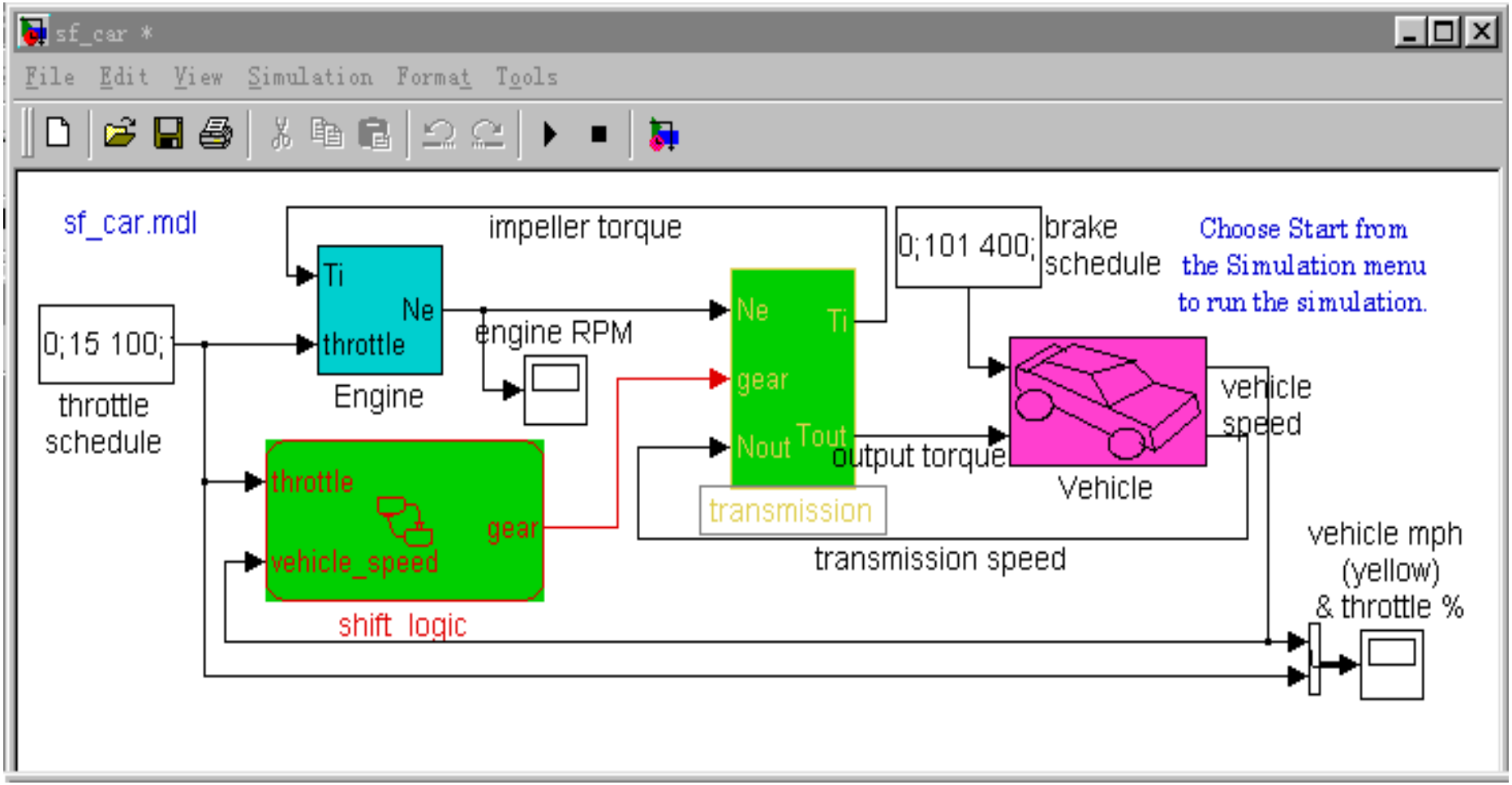
- 发动机转速闭环控制系统Closed Loop Engine Speed Control System



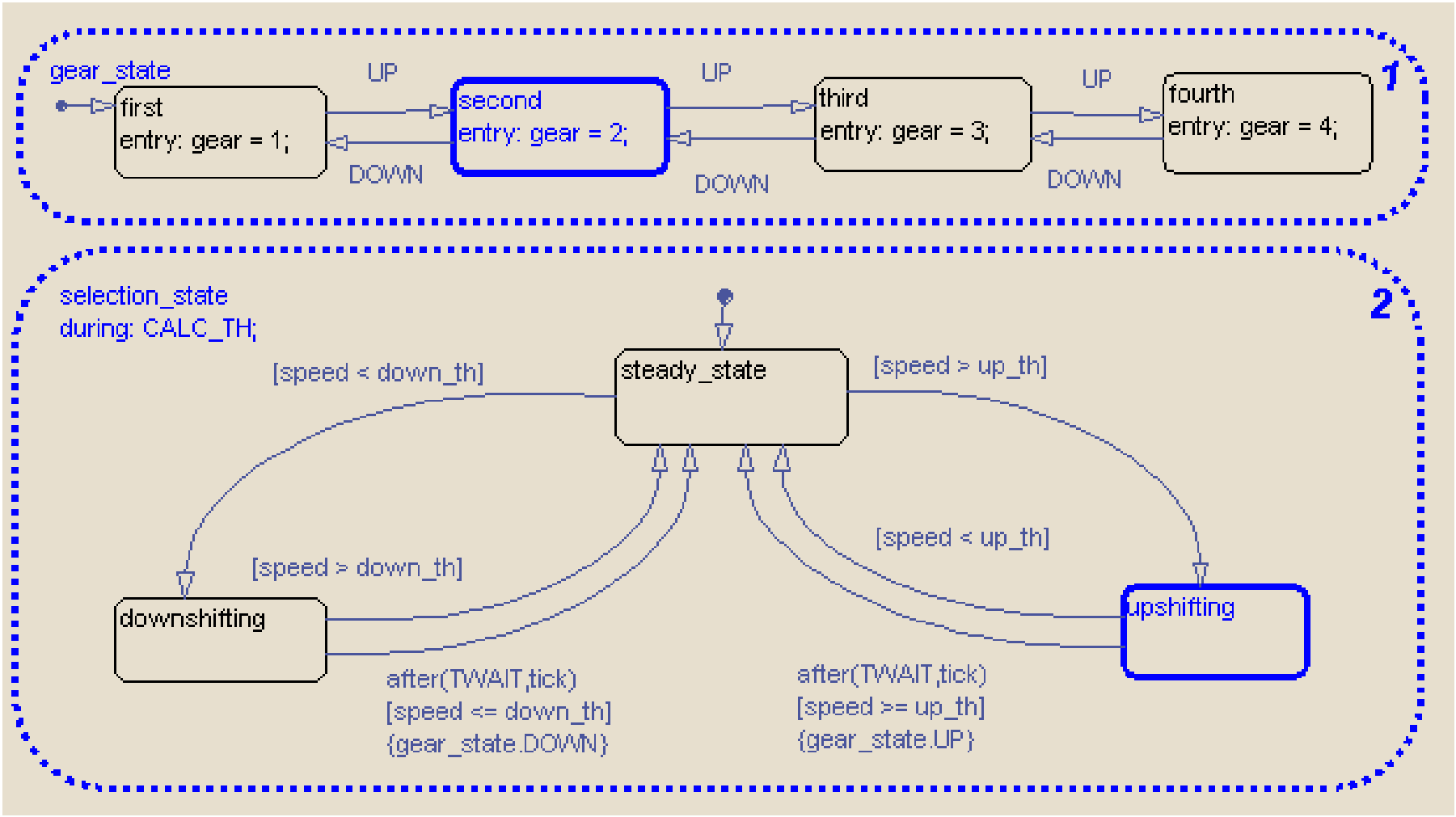

Closed-Loop Engine Speed Control
choose Start from the Simulation menu to run
valve timing
edge180
N
crank speed (rad/sec)
1
speed set point
Desired rpm
Throttle Ang.
N
Controller
Throttle Ang.
Mass Airflow Rate
Engine Speed, N
Throttle & Manifold
1/s
Intake
mass(k)
mass(k+1)
trigger
Compression
Air Charge
Torque
N
Combustion
Load
drag torque
Teng
N
Tload
Vehicle Dynamics
rad/s to rpm
30/pi
Engine Speed (rpm)
throttle deg (purple)
load torque Nm
(yellow)

Stateflow仿真举例：变速箱控制系统
Gear Box Control System

(MATLAB6/demo/Stateflow/A.T.C.)

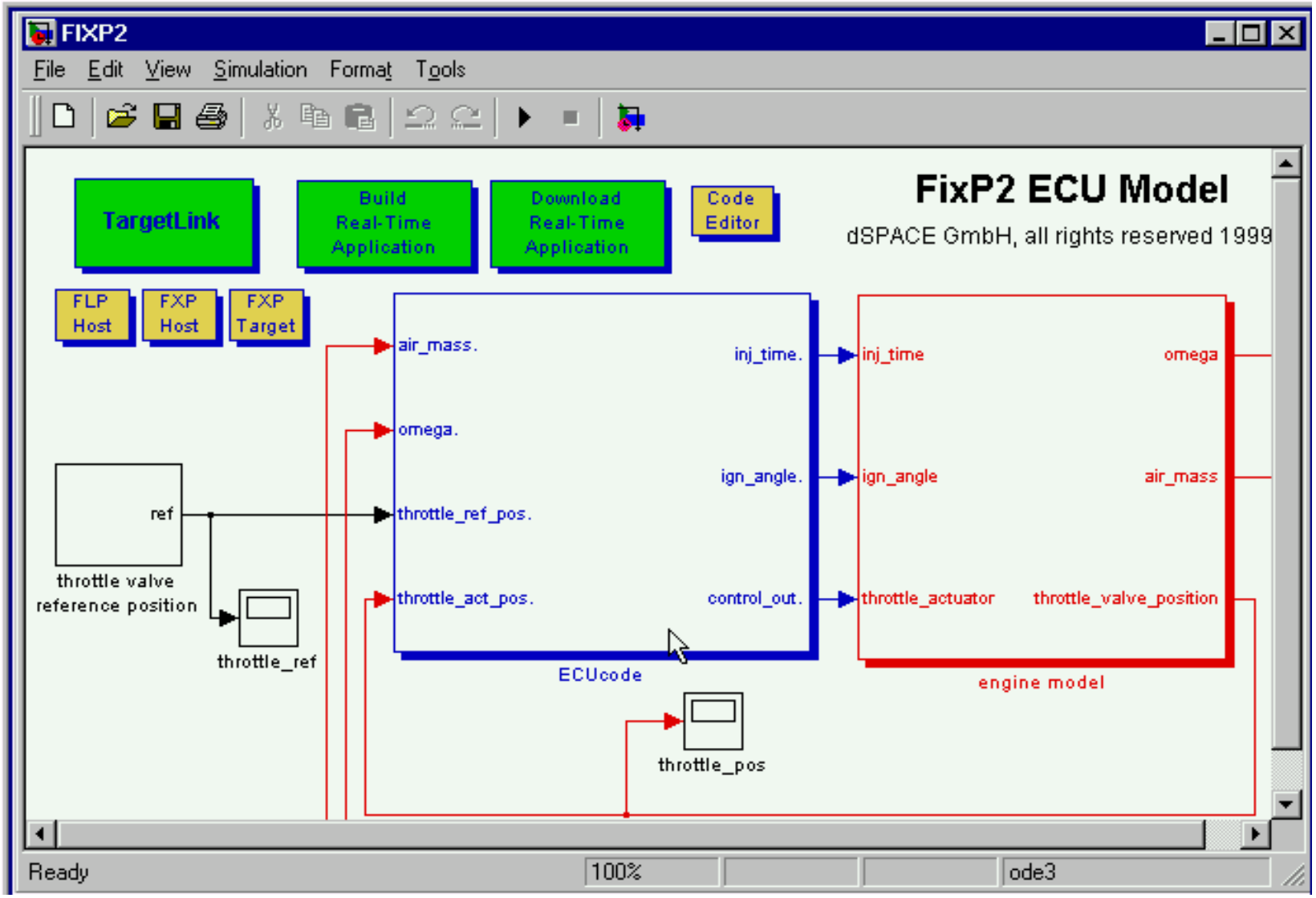


Stateflow仿真举例：变速箱控制系统
Gear Box Control System



(MATLAB6/demo/Stateflow/A.T.C.)

dSPACE开发环境功能简介 / Functions of dSPACE Platform



What is OSEK/VDX?

- OSEK/VDX is a joint project of the automotive industry. It aims at an industry standard for an open–ended architecture for distributed control units in vehicles.
- A real–time operating system, software interfaces and functions for communication and network management tasks are thus jointly specified.

OSEK/VDX的含义

The term OSEK means “Offene Systeme und deren Schnittstellen für die Elektronik im Kraft–fahrzeug” ---德语

(Open systems and the corresponding interfaces for automotive electronics).

---英语

The term VDX means Vehicle Distributed eXecutive.

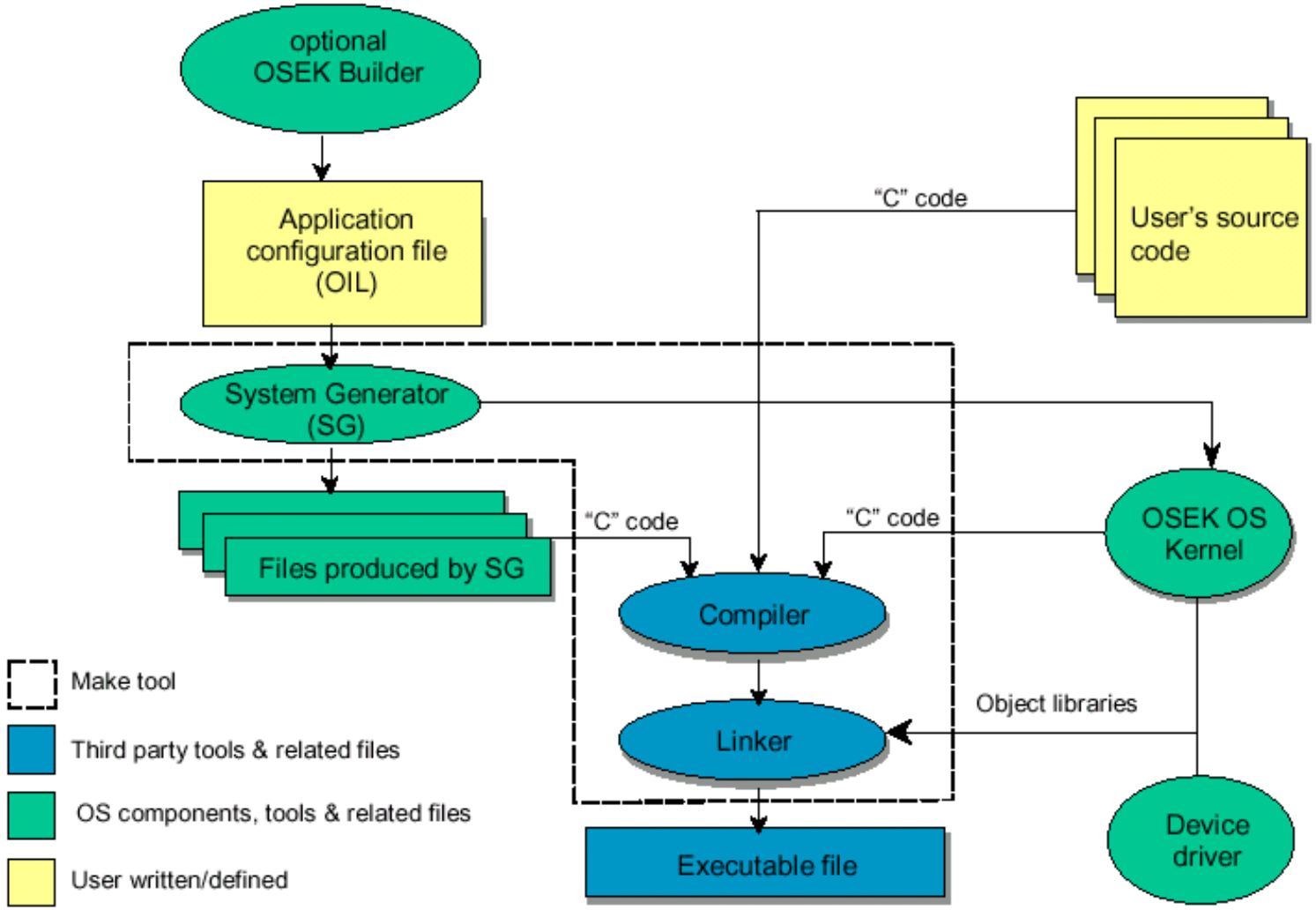
OSEK/VDX:

汽车电子的分布式开放系统及其接口标准

加入OSEK/VDX的部分著名企业

BMW（宝马） Fiat（菲亚特）
Denso（电装） Cummins（康明斯）
Delco（德科） Hewlett Packard（惠普）
ITT（国际电话电报） Lucas（卢卡斯）
Robert Bosch（博世） SGS Thomson（汤姆逊）
Siemens（西门子） Texas Instruments（德州仪器）
Daimler–Benz（戴姆勒–奔驰）
Volkswagen（大众） Volvo（沃尔沃）
Motorola（摩托罗拉） National（松下）
Philips（飞利浦）

OSEK环境下的开发过程
Software Developing Process



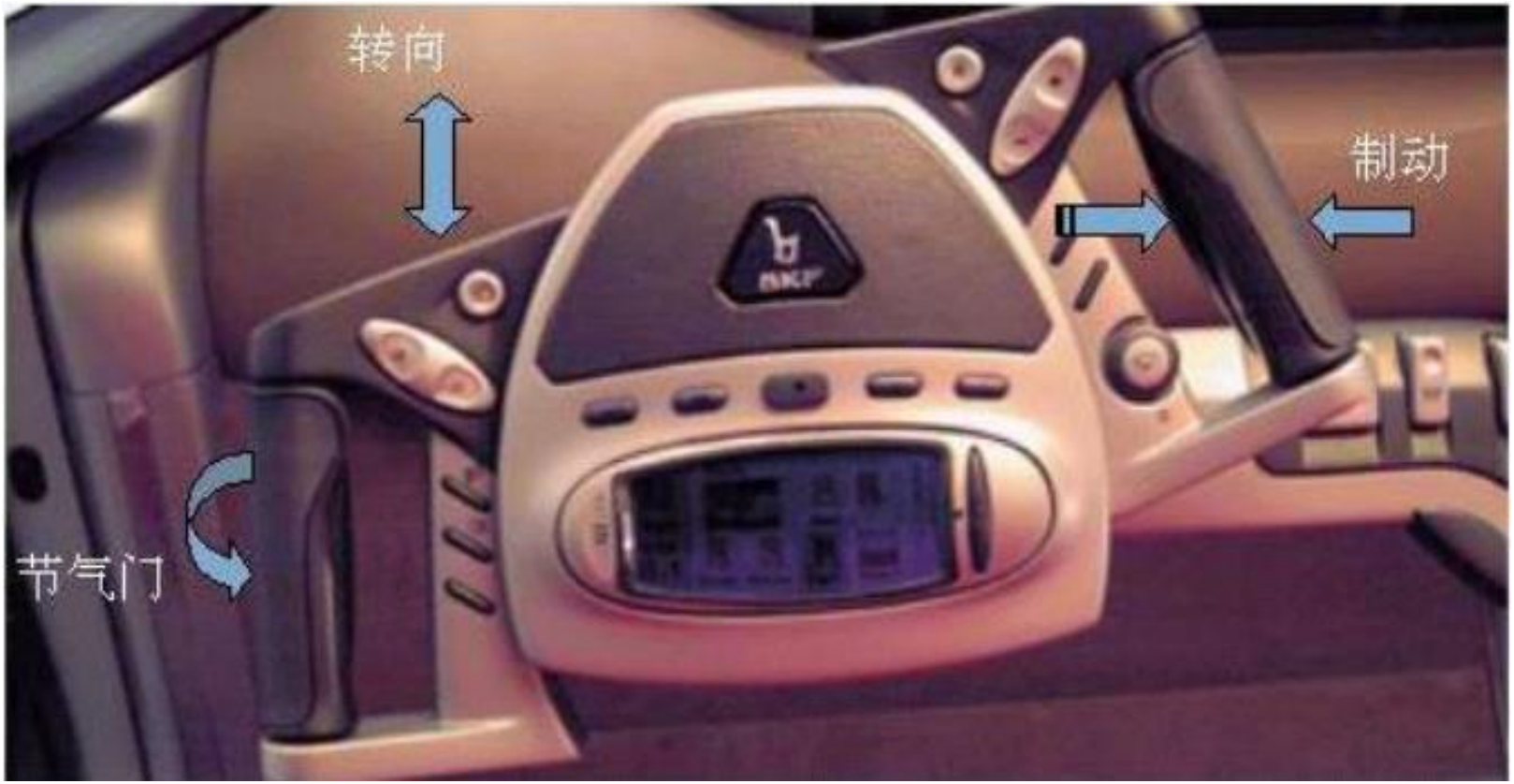
新型全电化燃料电池汽车技术
GM Autonomy

- 燃料电池发电系统，42伏电气系统，四轮电机驱动系统/Fuel cell engine，42V electrical system，four–wheel motor drive system
- 线传操控电子控制技术，没有脚踏板、仪表板和转向柱/X by wire without accelerator pedal、instrument panel and steering pole.
- 底盘和车身分开设计，车身命令通过网络接口连接底盘控制系统 /Chassis and bodywork designs are separate with standard Mechanical and Network Interface

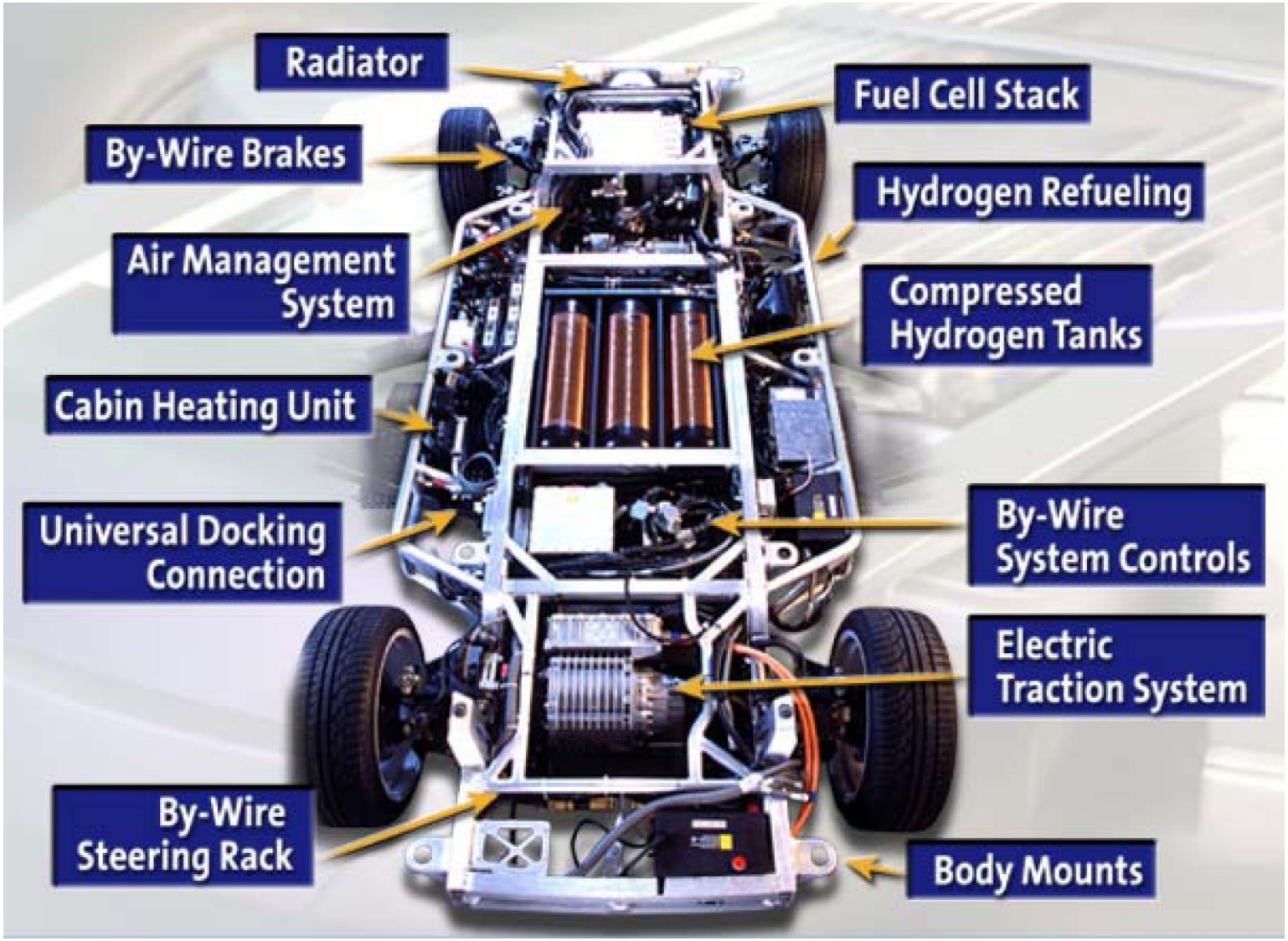




新型燃料电池混合动力技术
Hy–wire Driver Interface



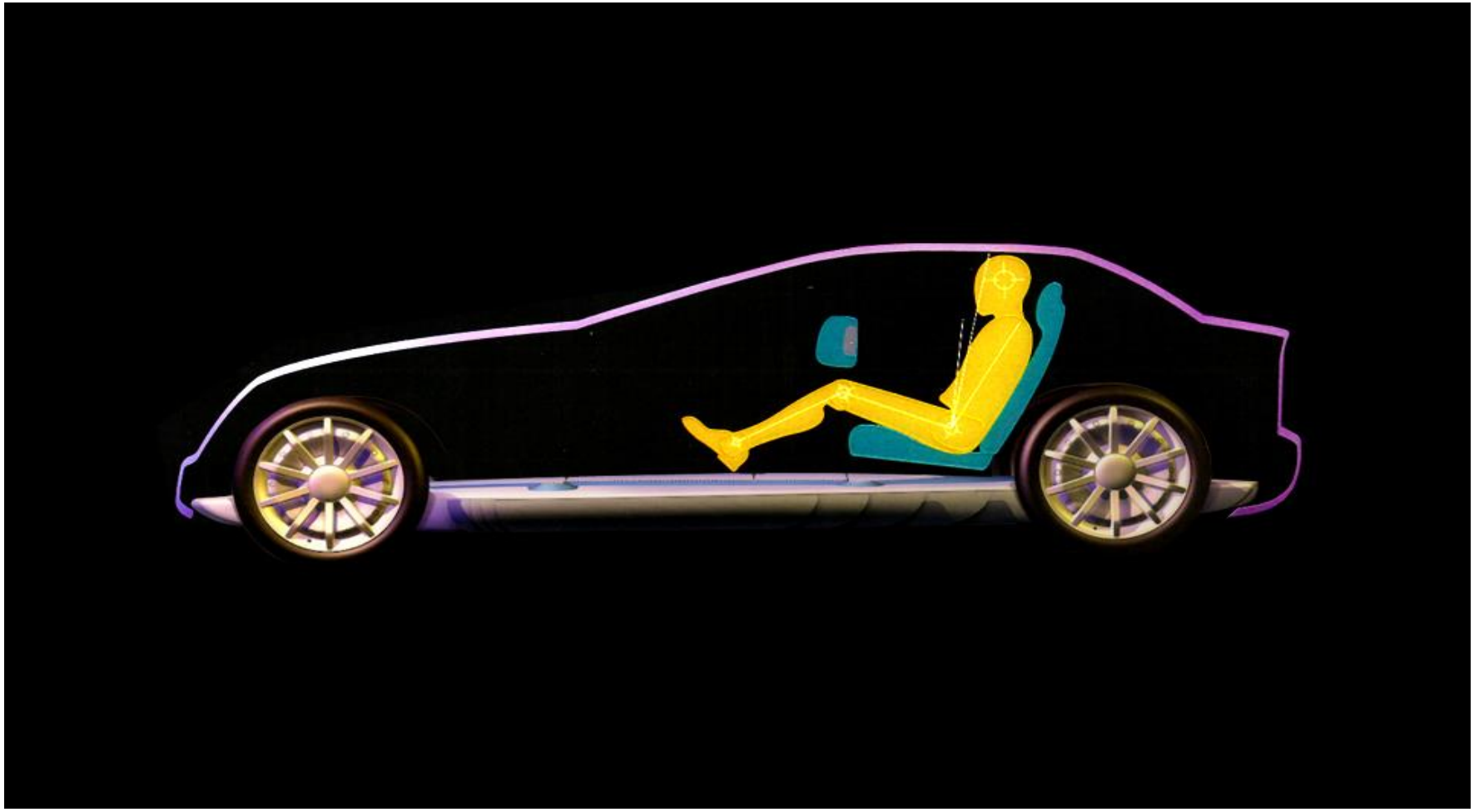
新型燃料电池混合动力技术
Hy–wire Chassis Structure



新型燃料电池混合动力技术
Hy–wire Assembly of Car body



GM: “Hy–Wire” Act
Change different style with same chassis



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/Characteristics of AECS：Strong Independence

2 汽车电子的三个阶段

/Development Phases of AECS

3汽车电子的基本框架

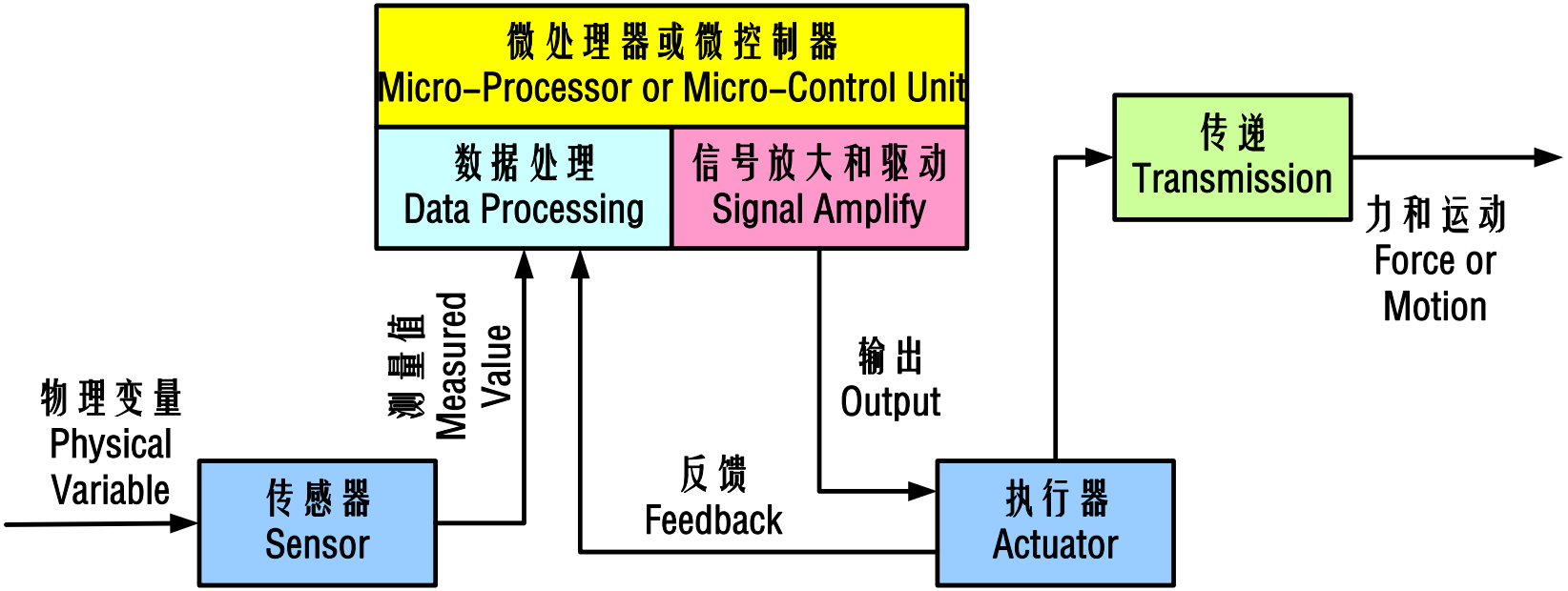
/Basic Framework of AE Control Systems

4汽车电子的开发流程

/Development Process of AECS

§3 汽车电子的基本框架：硬件框架

- 汽车电控系统的基本组成：/Basic Components of Automotive Control Systems
  - 传感器、执行器和控制器构成汽车电子控制系统的基本要素/The Basic element of AECs are: Sensors, Actuators, and Controller.



§3 汽车电子的基本框架：硬件框架

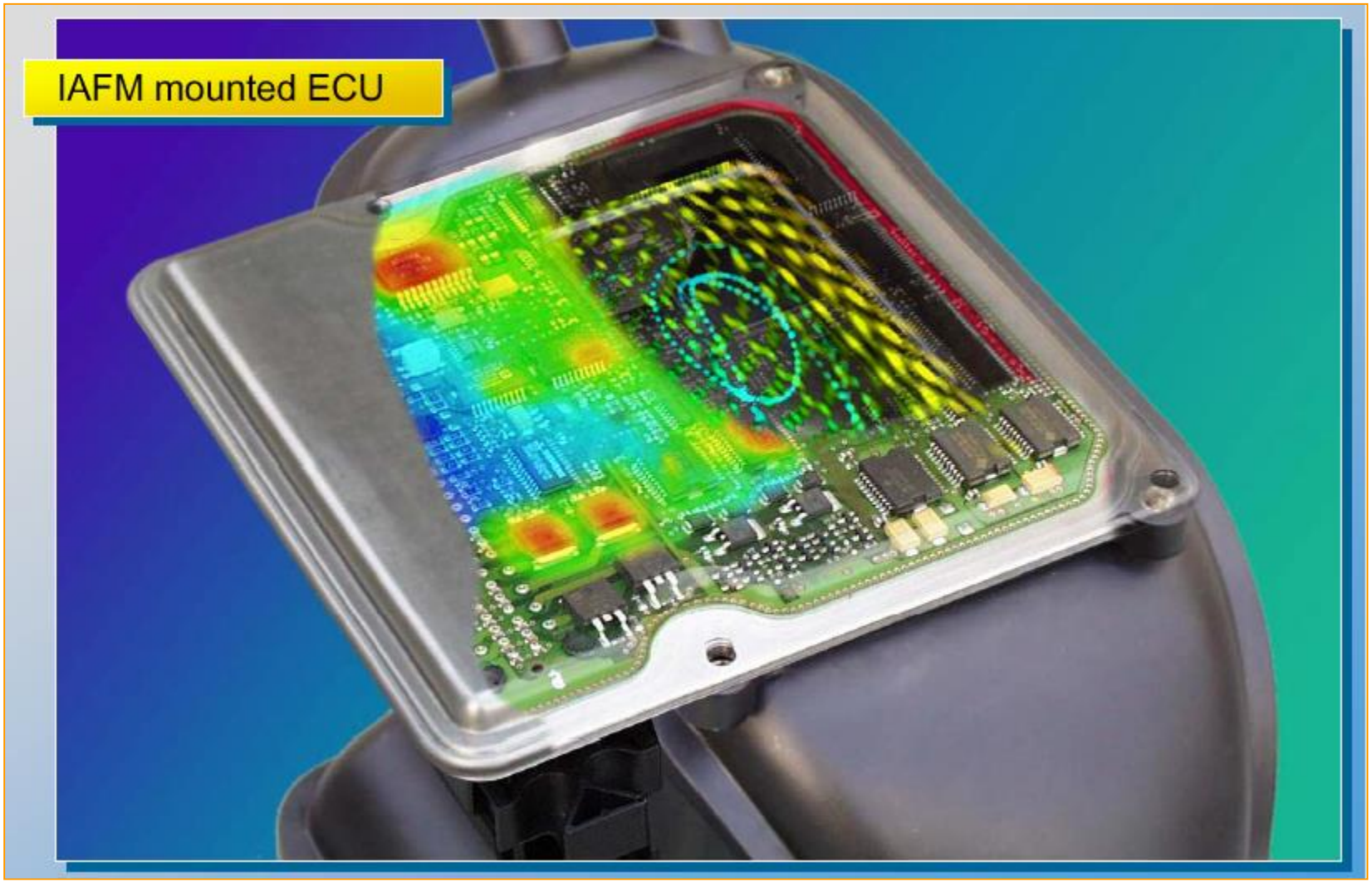


§3 汽车电子的基本框架：硬件框架

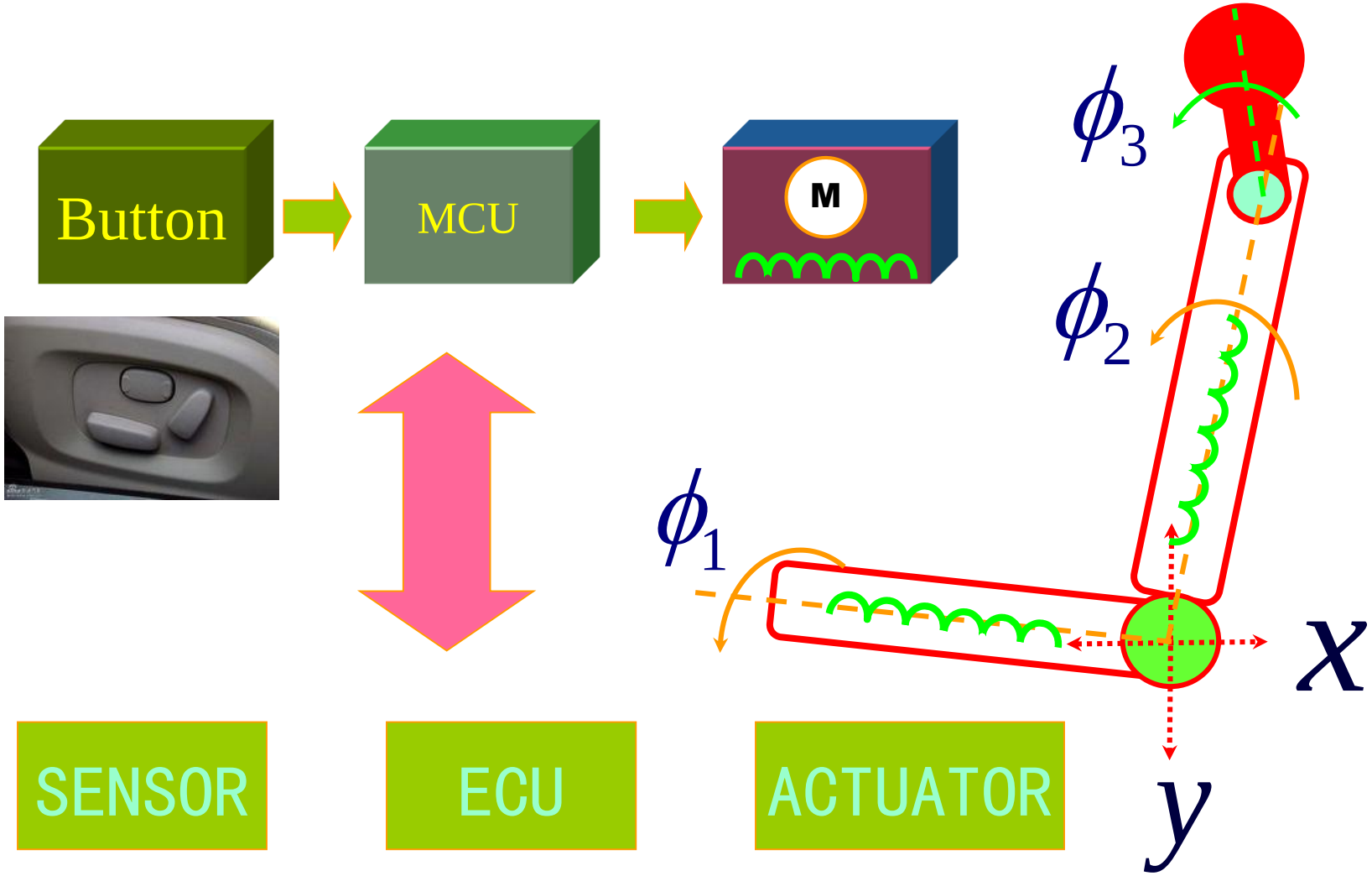
电子控制单元（ECU）主要包括如下五部分/The ECU Mainly Consists of 5 Parts:

- 传感器接口/ Sensors → input signals：feedback and feedforward ；
- 执行器驱动/ Actuators → output signals：control objects；
- 数字核心/ Digital core – >signals processing：control arithmetic
- 电源系统/ Power system：12V，24V，42V
- 通讯系统/ Communication system：on–board network

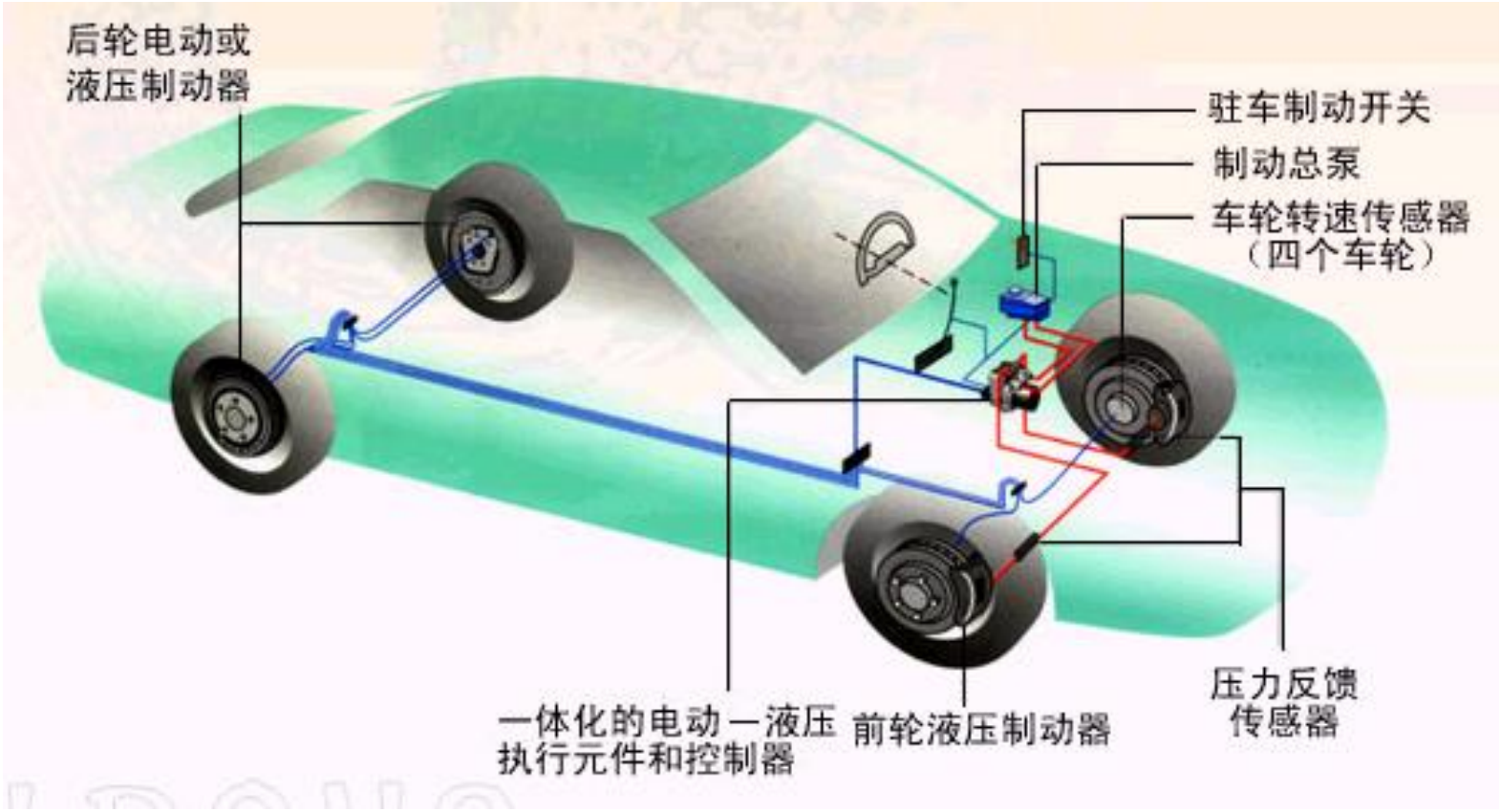
§3 汽车电子的基本框架：硬件框架



§3 汽车电子的基本框架：硬件框架

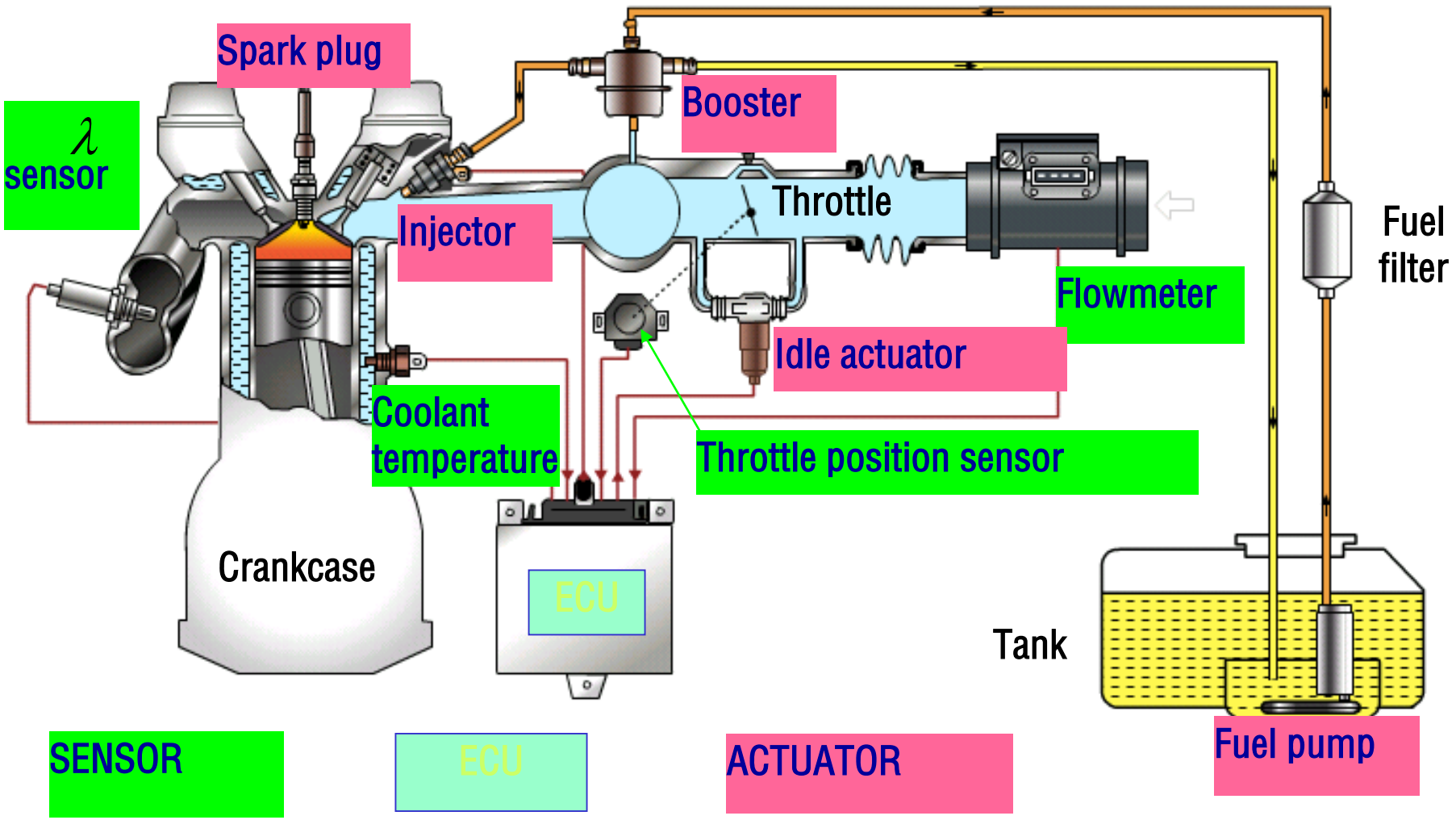


§3 汽车电子的基本框架：硬件框架

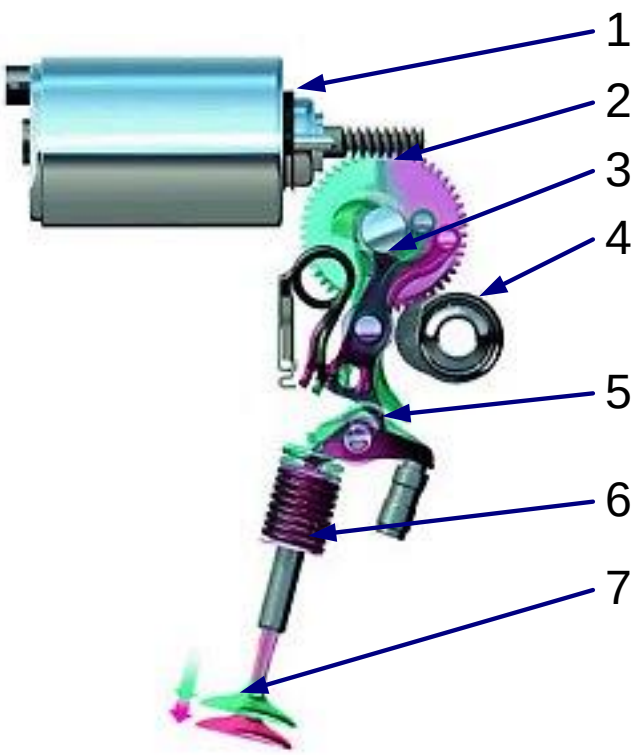


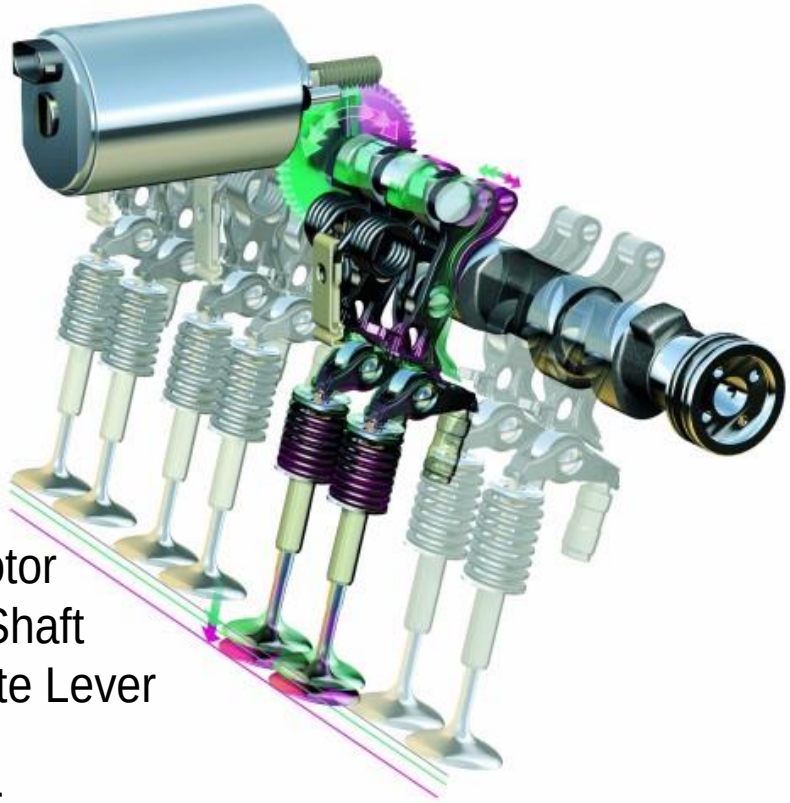


§3 汽车电子的基本框架：硬件框架



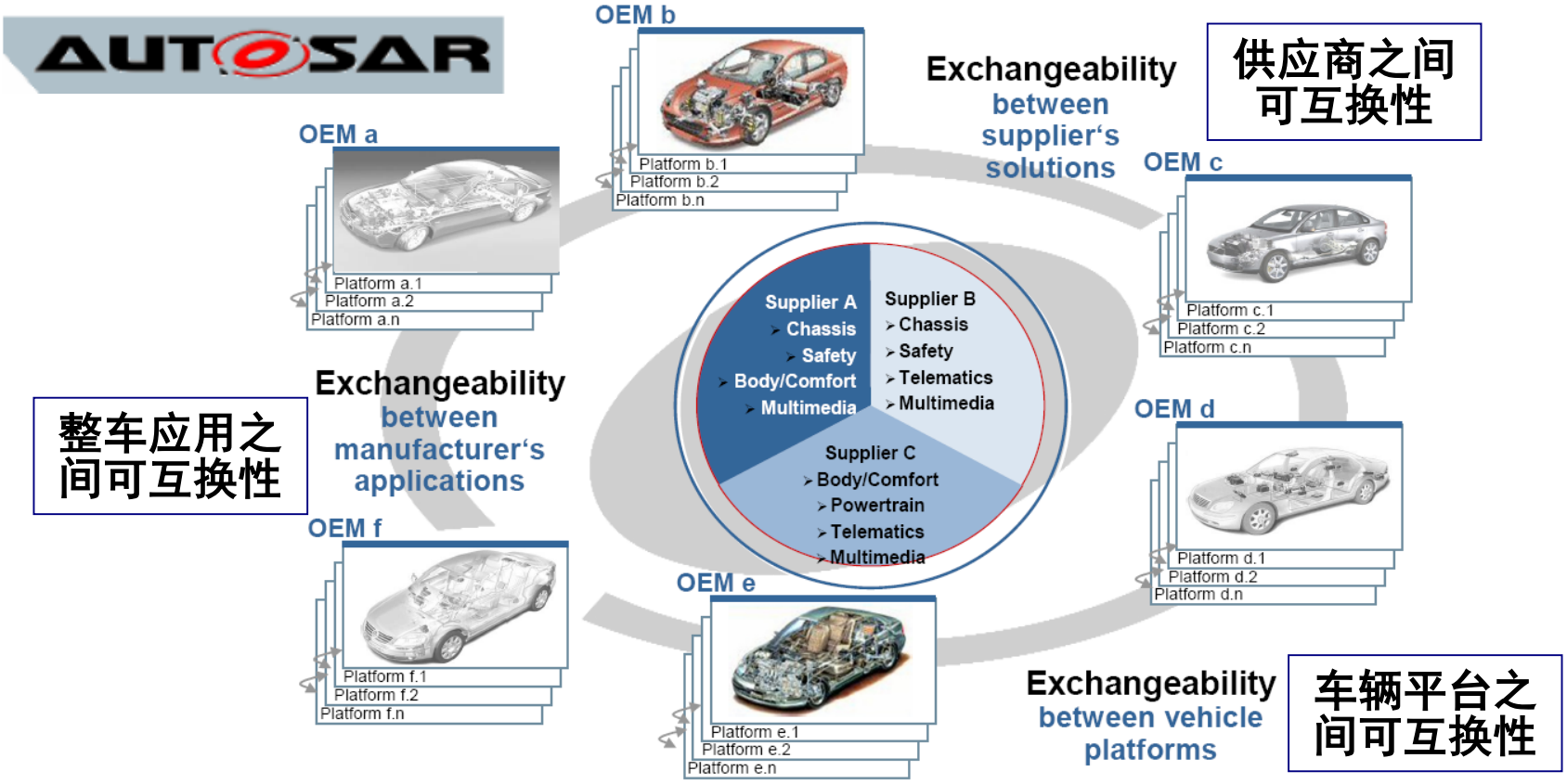
•可变气门升程控制系统/ AECs Example: Valvetronic system



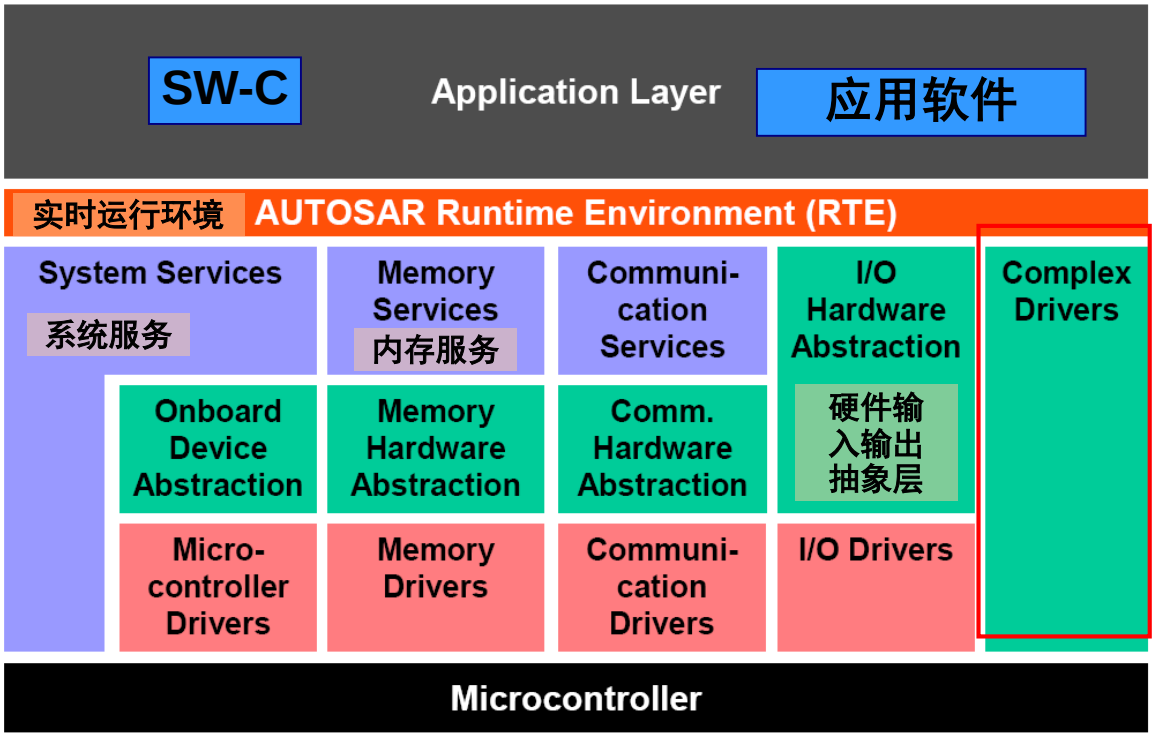


1 Electric Motor
2 Eccentric Shaft
3 Intermediate Lever
4 Camshaft
5 Cam Lever
6 Valve Spring
7 Inlet Valve

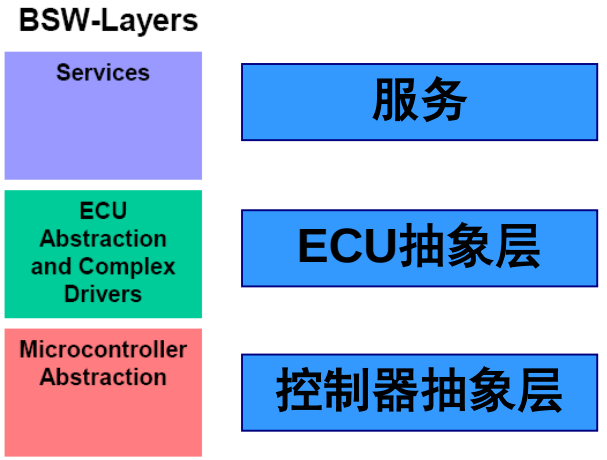
- AUTOSAR：AUTomotive Open System ARchitecture



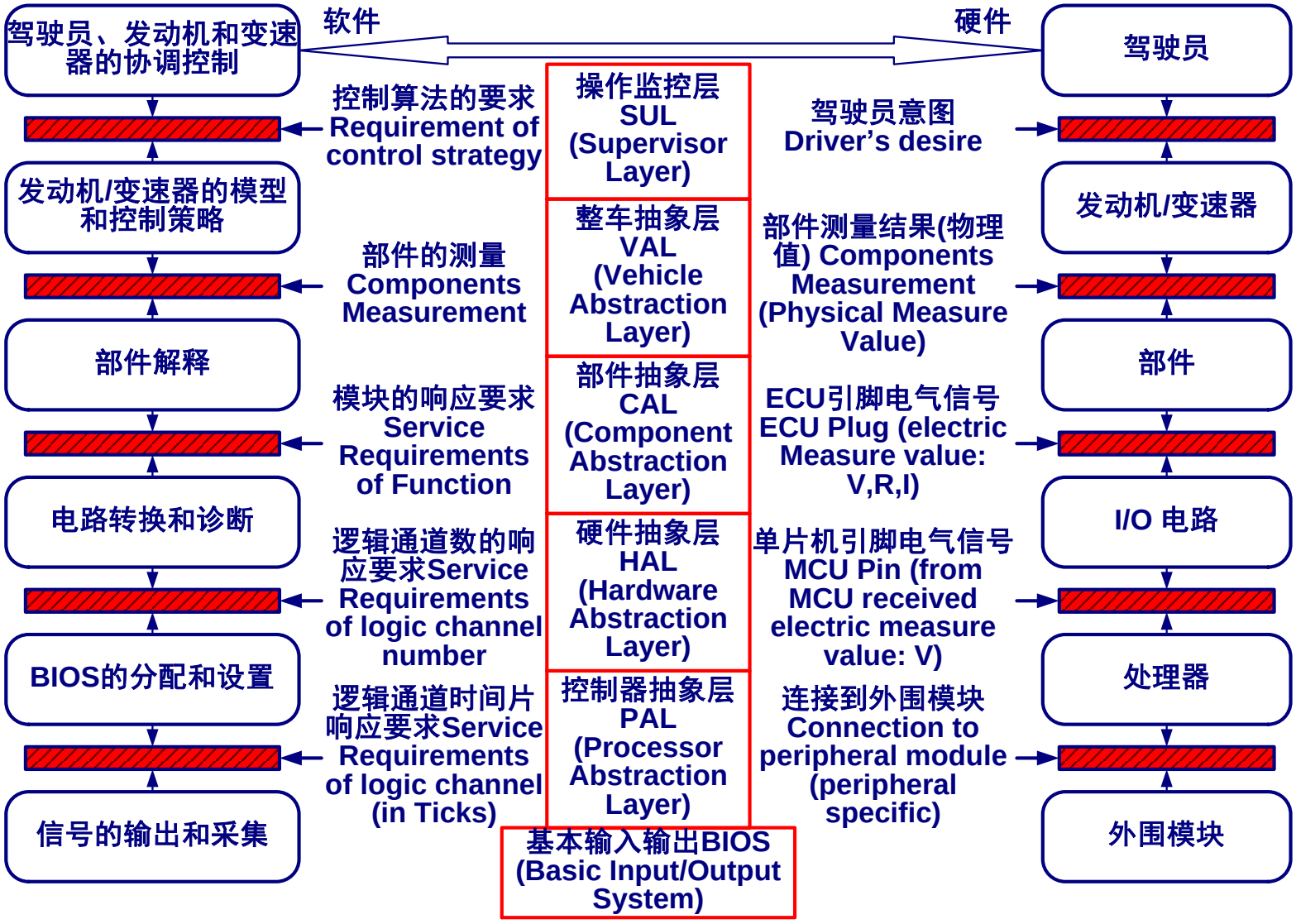
●AUTOSAR 的层次



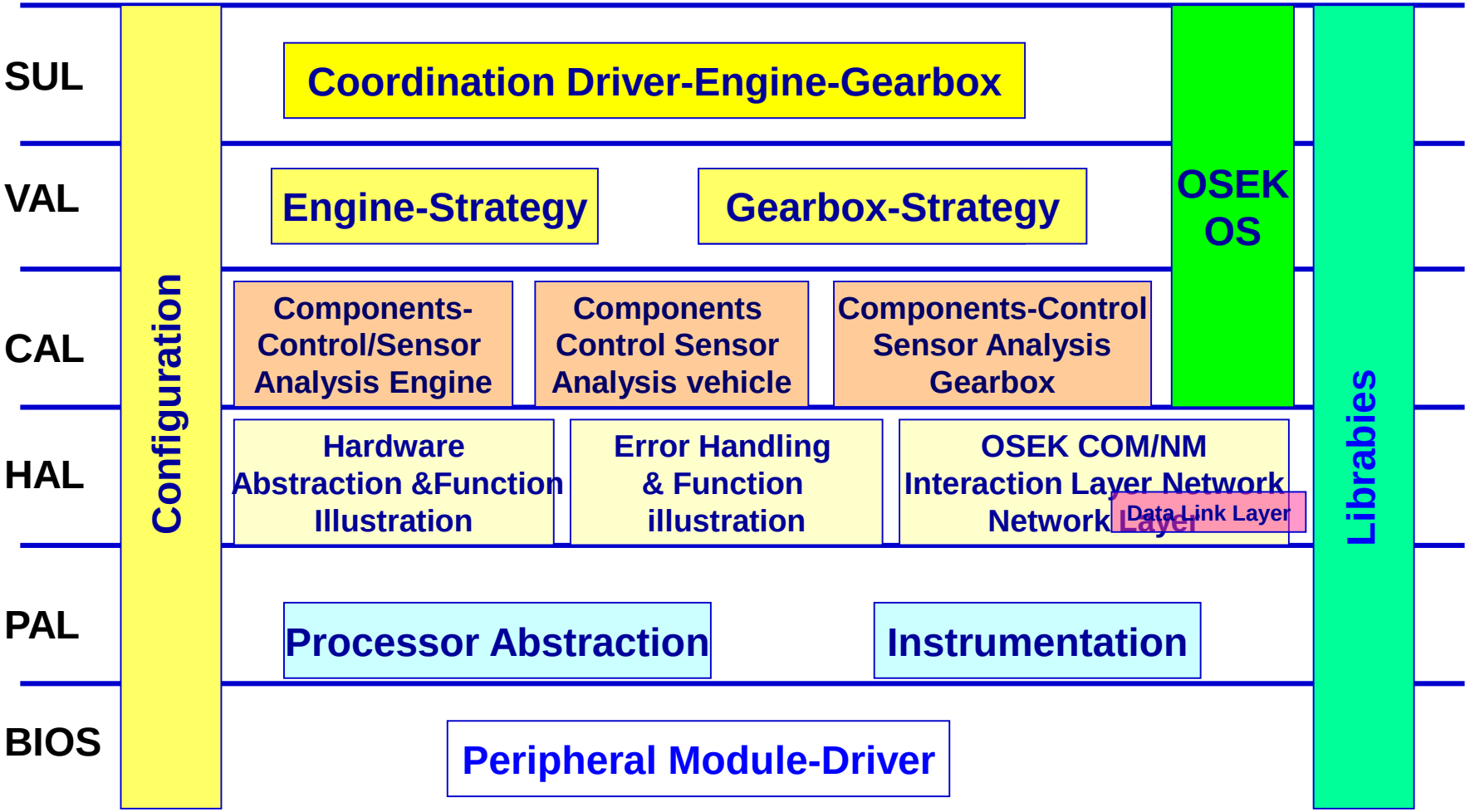
复杂驱动（Complex Drivers，间称CD）指发动机控制中的曲轴同步，点火和喷射控制，以及电磁阀驱动等。



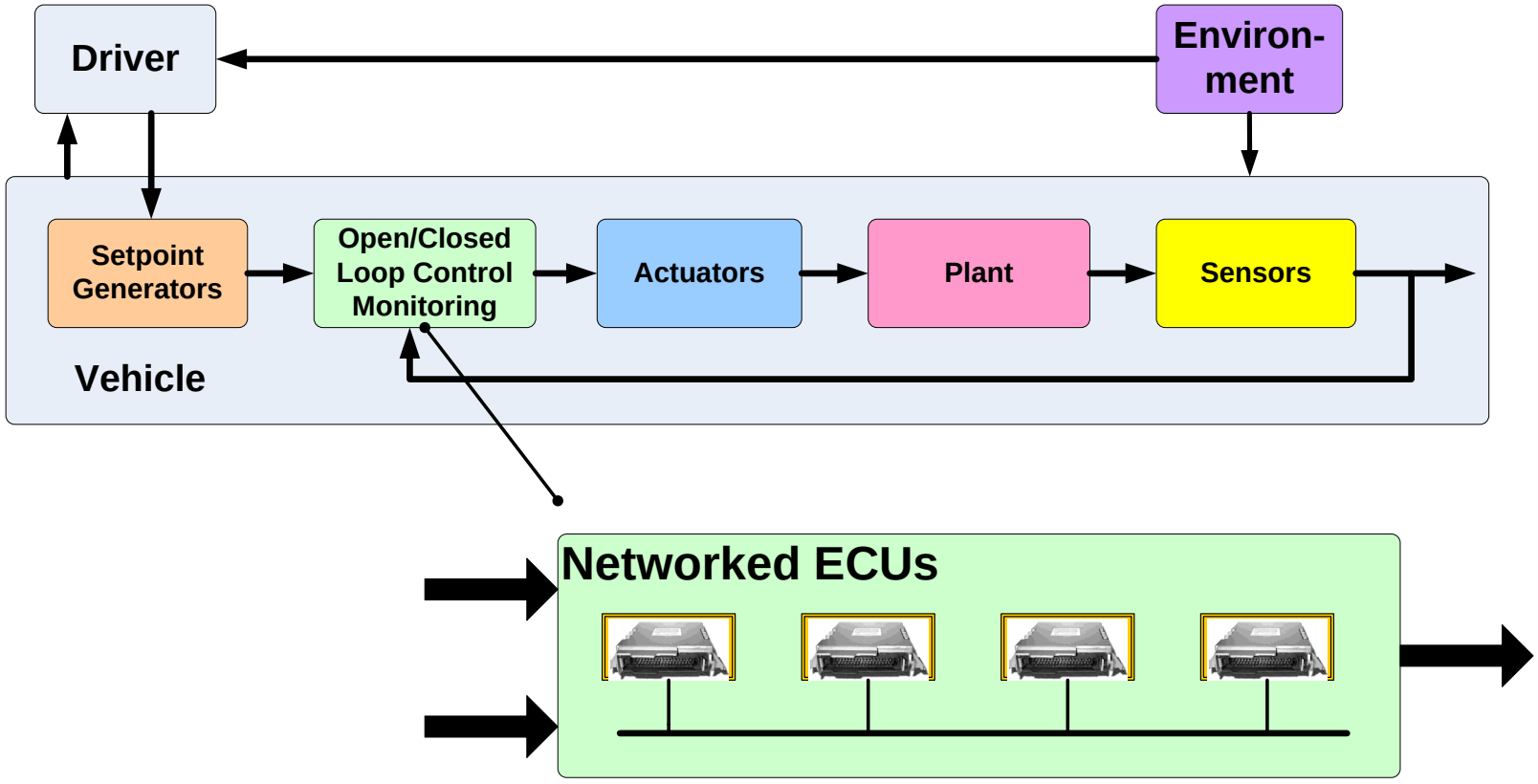
§3 汽车电子的基本框架：软件框架



§3 汽车电子的基本框架：软件框架



- 驾驶员–控制系统–车辆–环境构成闭环控制系统/Driver–AEs–Vehicle–Environment Consists Closed Loop Control System



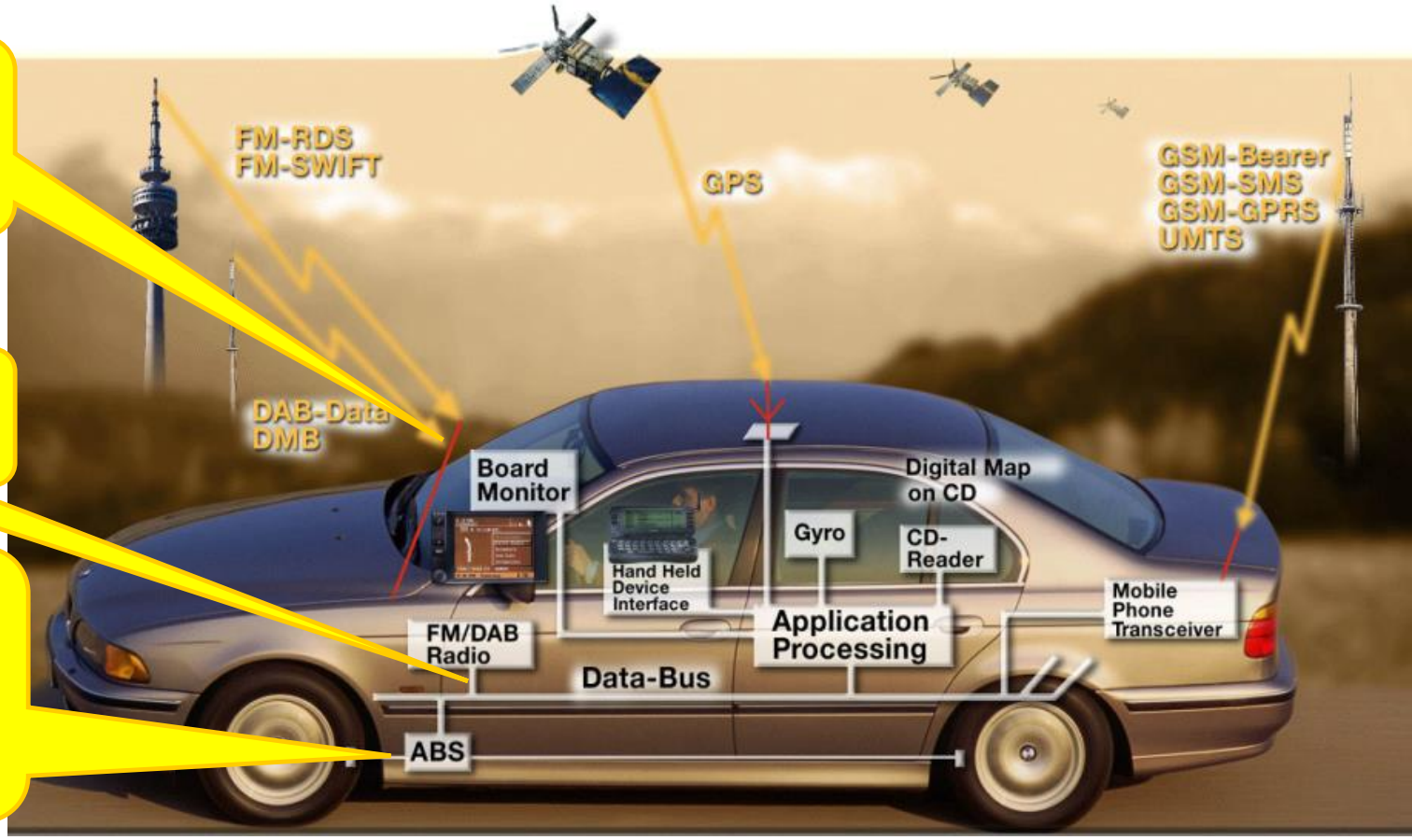
§3 汽车电子的基本框架：网络架构

- 零部件的电子化建设,车载网络/多媒体系统/Construction of Electrical System, onboard network/ multimedia system

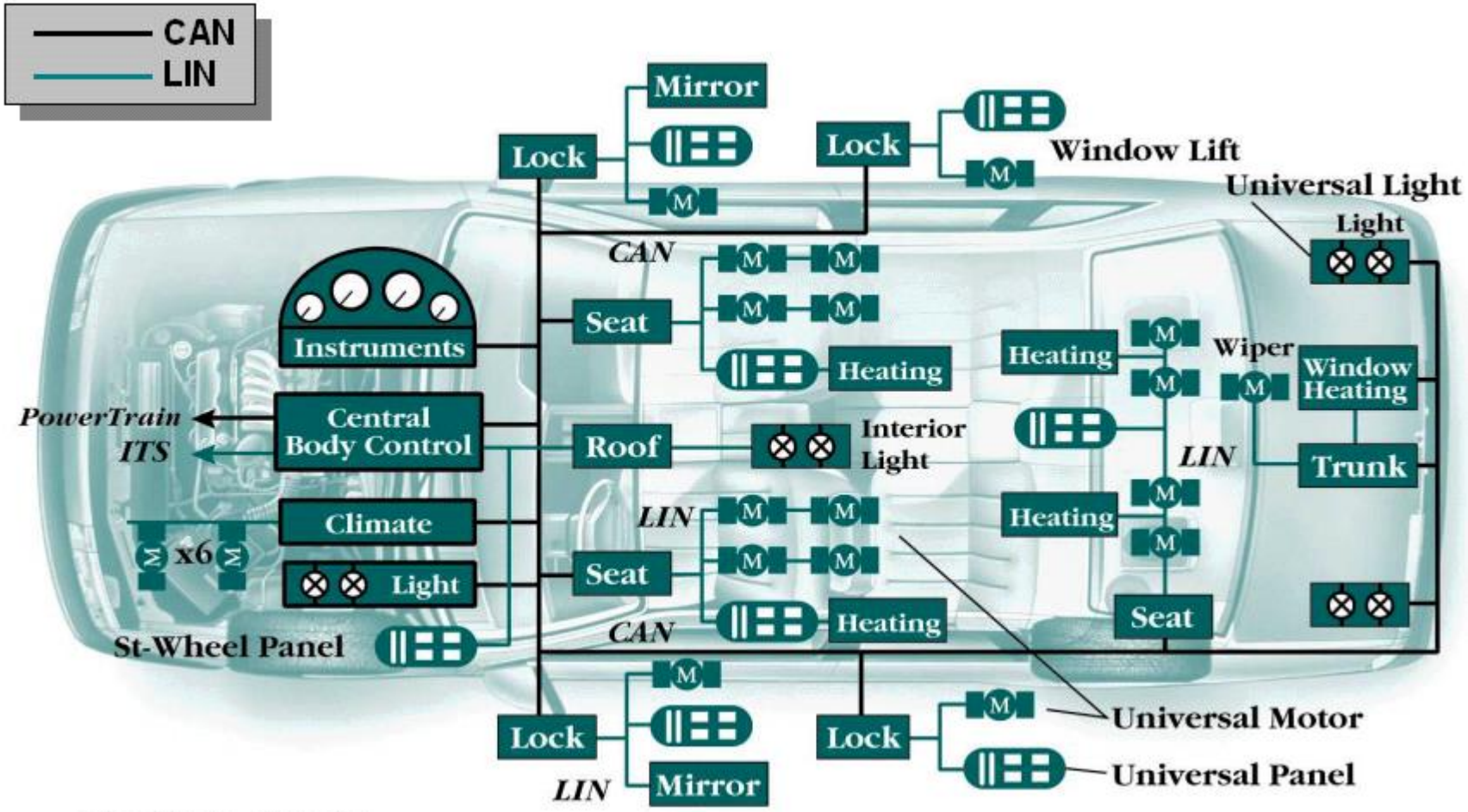
Inter Vehicle network

Inner Vehicle network

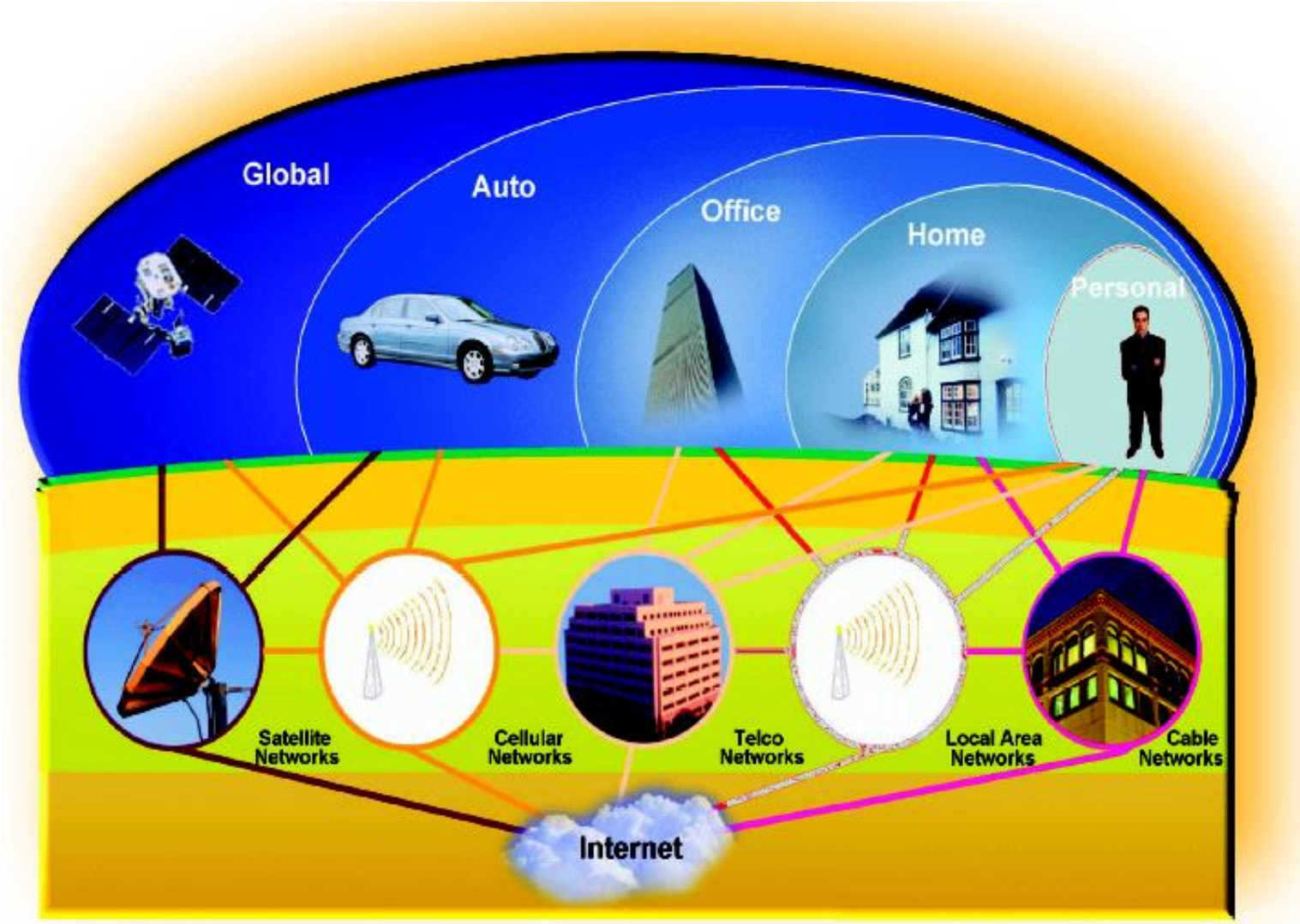
Control subsystem & Components



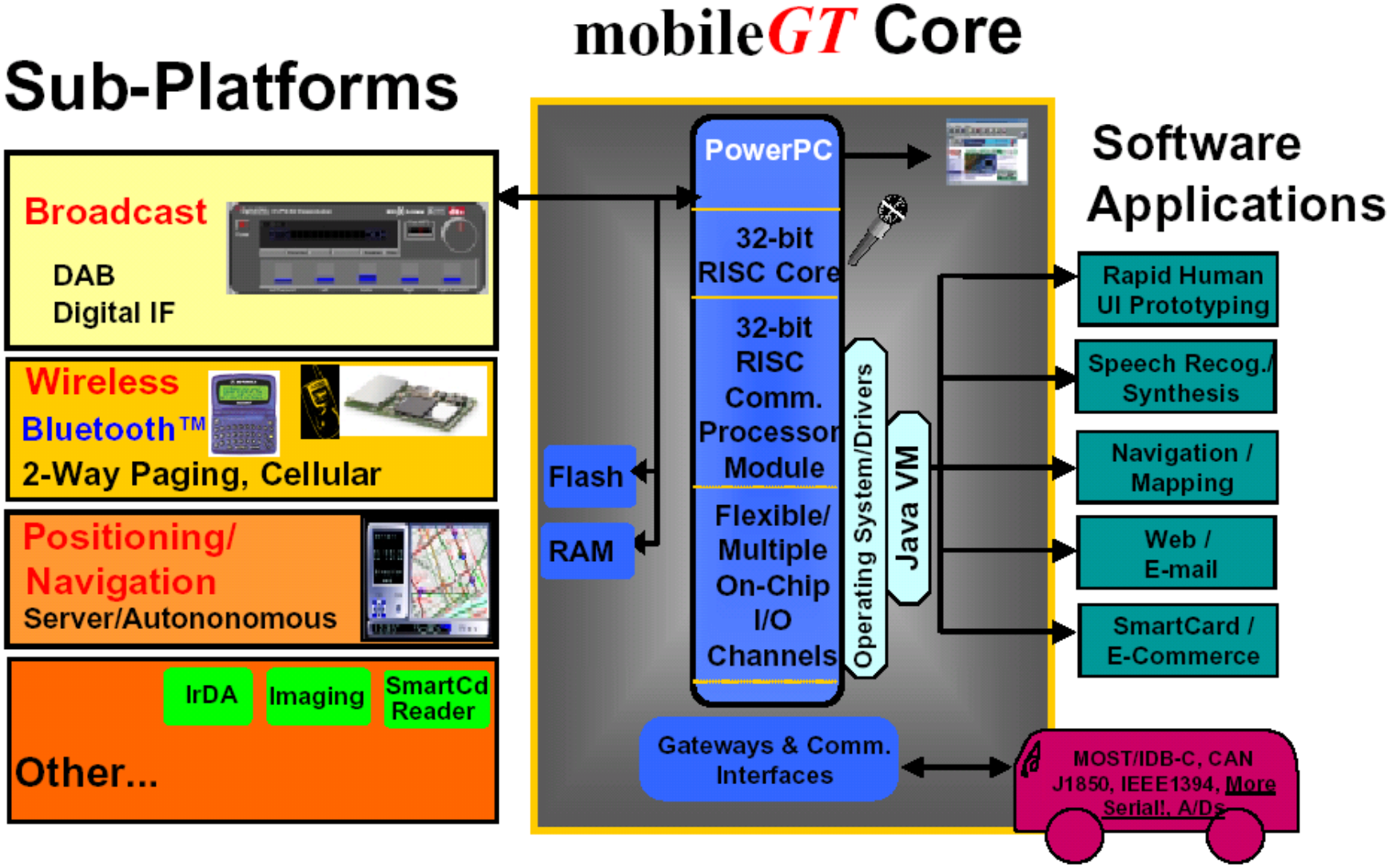
§3 汽车电子的基本框架：网络架构



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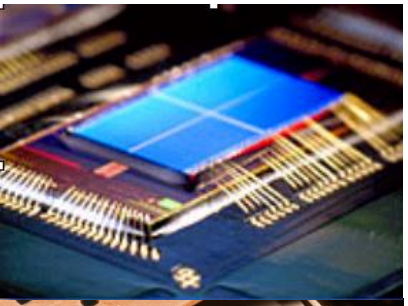


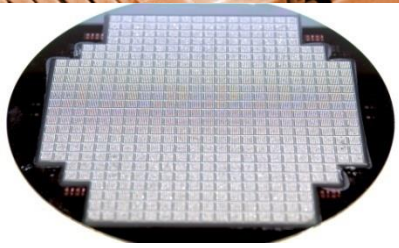
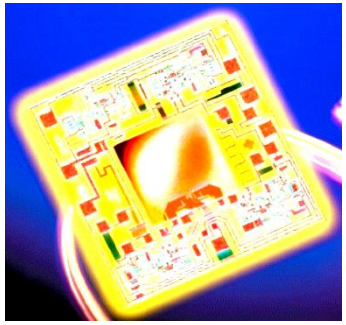
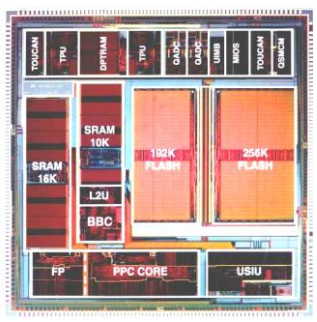
- 汽车半导体/Automotive Semiconductor

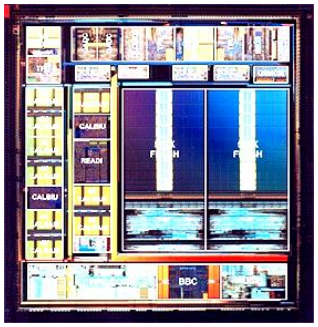
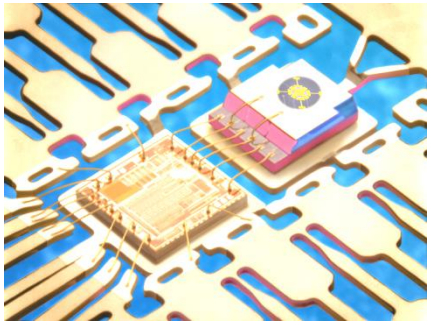
产品升级/ Product Improvement → 适应新变化/Adapt New Requirements

开发工具 /Development tools → 培训和应用/ Training and Application

芯片设计生产/Chips design and Produce → 汽车市场需求/Automotive Market Requirements

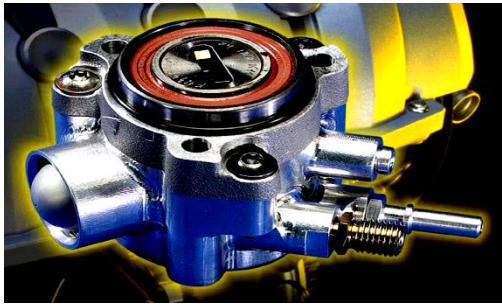


§3 汽车电子的基本框架：核心技术

执行器执需要综合机－液－电－磁等多领域的技术，是设计、制造和加工要求都很高的尖端产品/ The actuators are the key components and parts of the AEs. The R&D of those components needs the deepen inter–disciplinary knowledge–Mechanic, Hydraulic, Electric and Magnetic. They needs high accuracy and precise technique in design and manufacture.













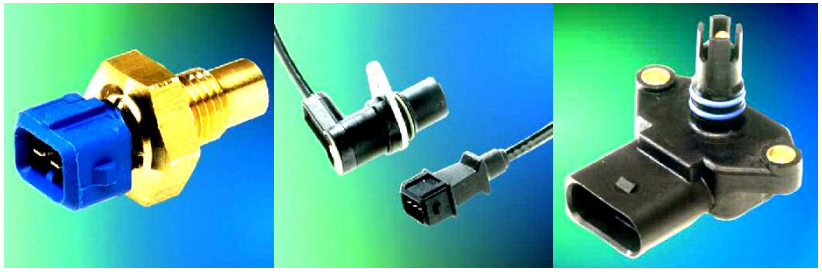
§3 汽车电子的基本框架：核心技术



阶段3：自主研发，掌握技术含量高、更加关键的新产品/Step 3: Independently R&D new products with complexity key technology

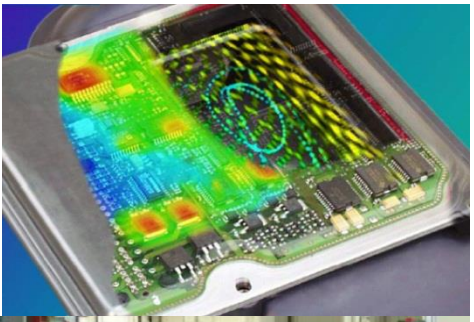


阶段2：大批量、低成本、专业化生产，消化关键技术/Step 2：large-volume、low cost、specialization, familiar to the key technologies of the product.



阶段1：以低技术含量、低价加工对象为切入点，进入国际供应体系/Step 1：the start point to enter the supply system of the world is to mass produce the components with less amount of technology

§3 汽车电子的基本框架：核心技术



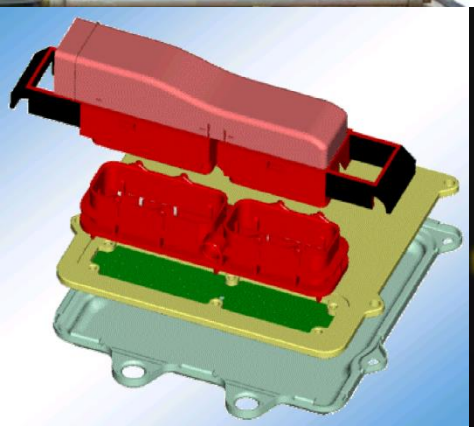


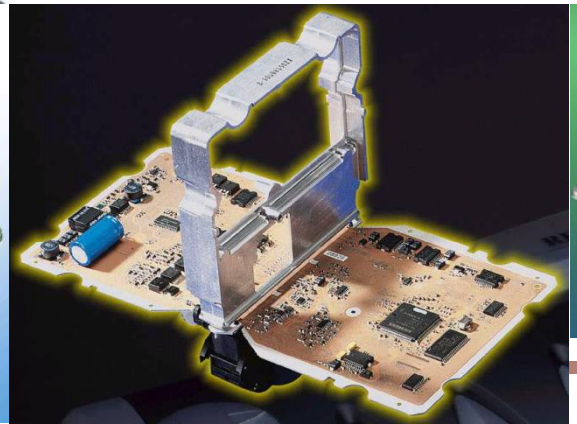
全球电子元件供应体系/Global purchase for Automotive Semiconductor

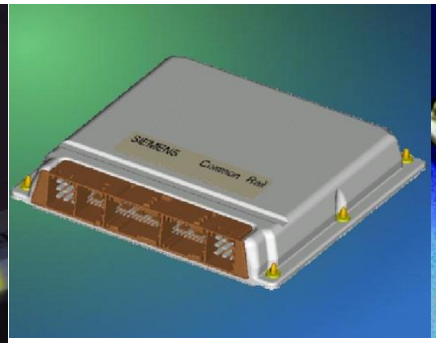
开发硬件和软件/Design in Hardware and Software

系统匹配与标定/ Optimization and Calibration

大批量、低成本规模化生产/Mass production in low cost

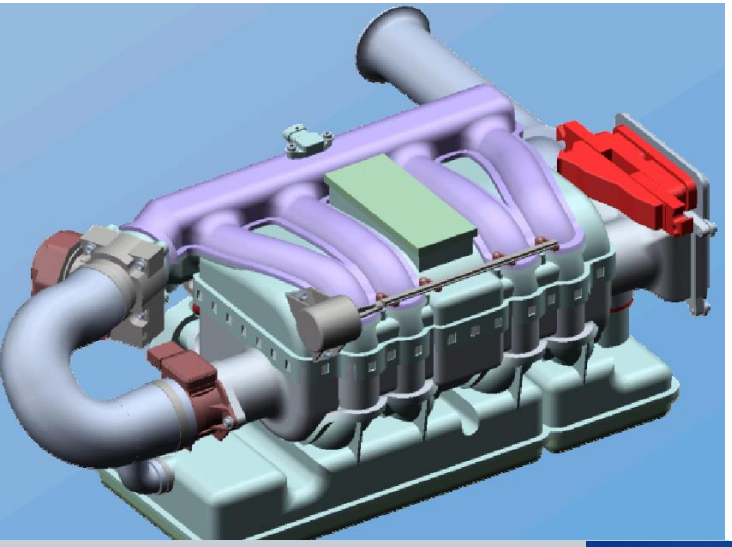




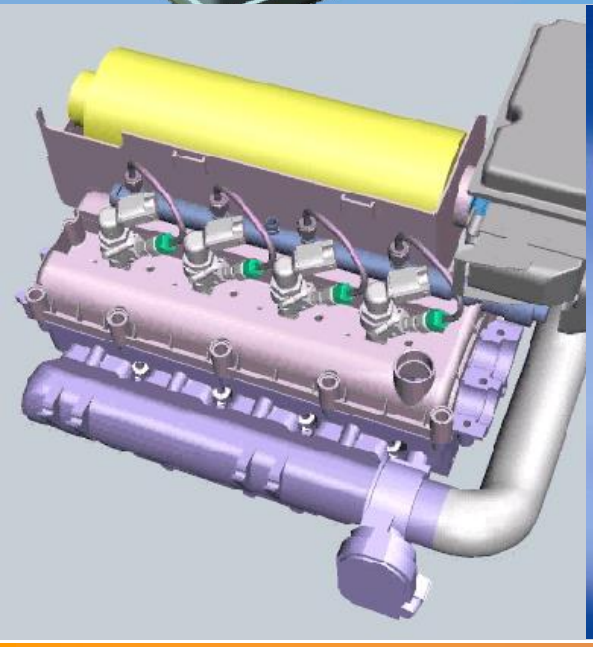




§3 汽车电子的基本框架：核心技术



- 各种部件控制系统－汽车零部件升级和换代的主战场/Control systems of all kinds of parts––the main battlefield of upgrading automotive parts







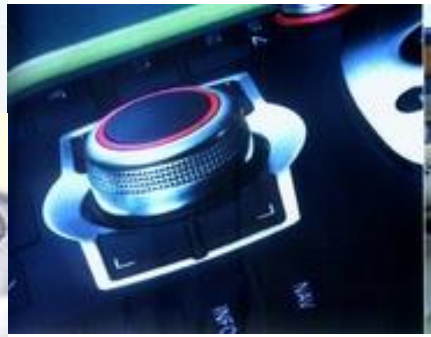
§3 汽车电子的基本框架：核心技术







- 整车网络属于信息产业在移动车辆上的延伸。结合国家的信息技术发展来确定整车网络的发展和方向./In-vehicle network is the extent of information industry in moveable vehicles. The development of in-vehicle network should joint into the system of National information technology.





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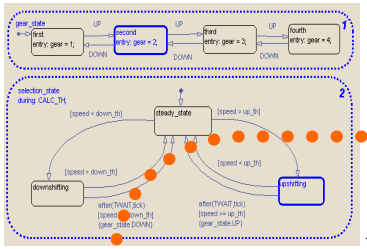
2 汽车电子的三个阶段
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4汽车电子的开发流程
/Development Process of AECS

电控系统V型开发模式基本流程

功能定制





系统集成标定

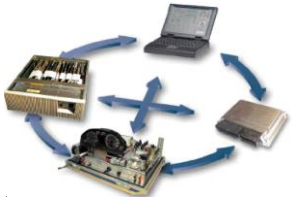
原型设计

系统校验



dSPACE RCP

技术实现



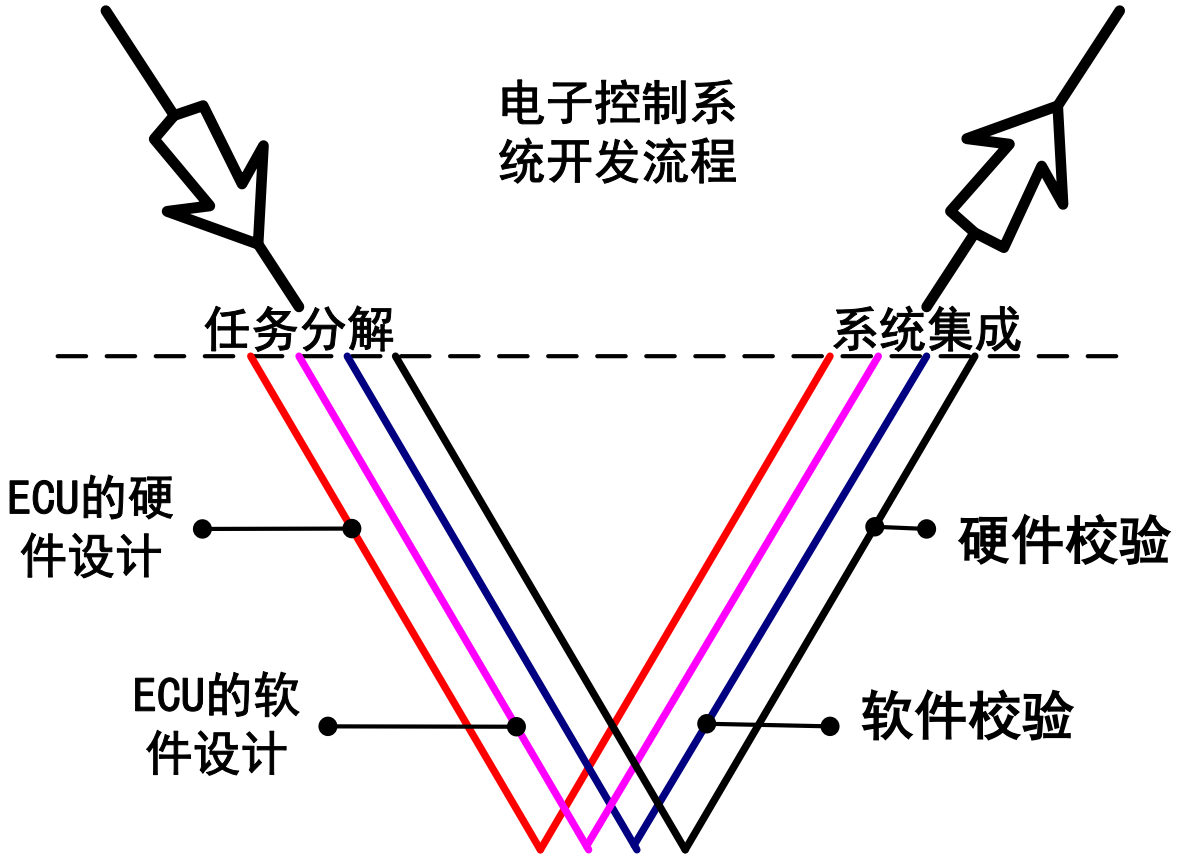
dSPACE HIL





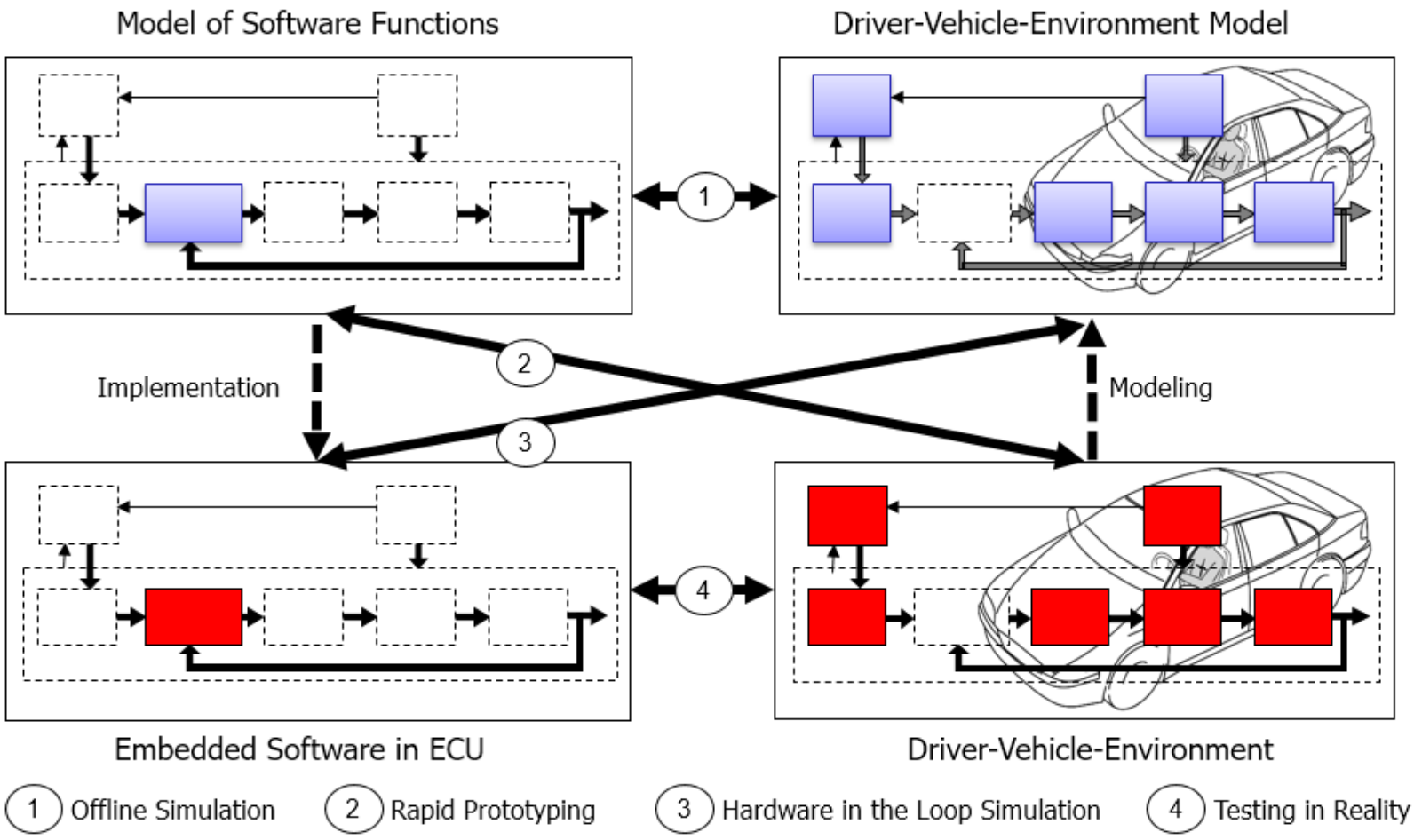
TargetLink/Codewarrior/RTOS/Protel DXP

电控系统V型开发模式基本流程

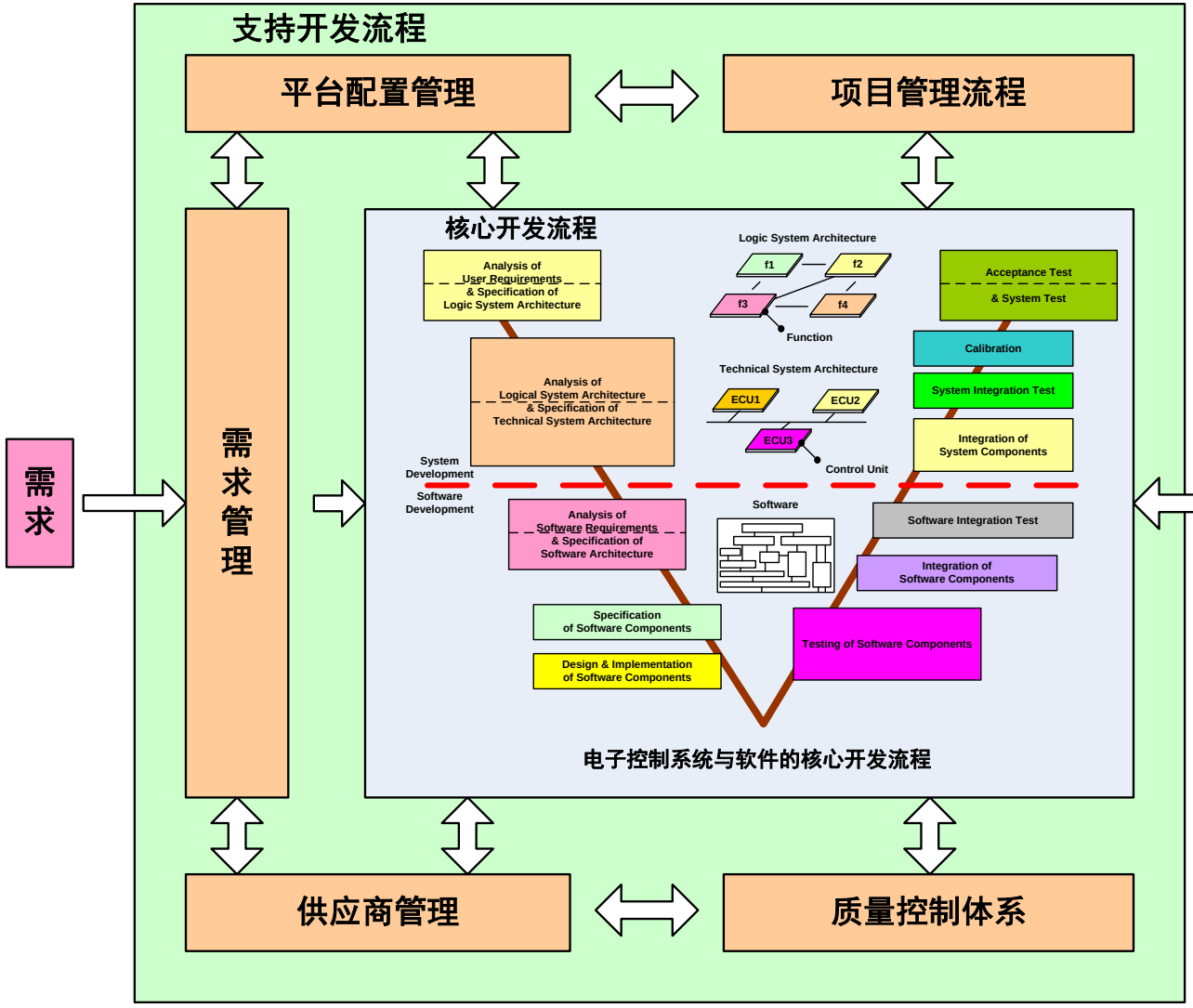


（2）汽车电子软件的核心开发流程 /Core Process of Software Development

电控系统V型开发模式基本流程



项目管理框架

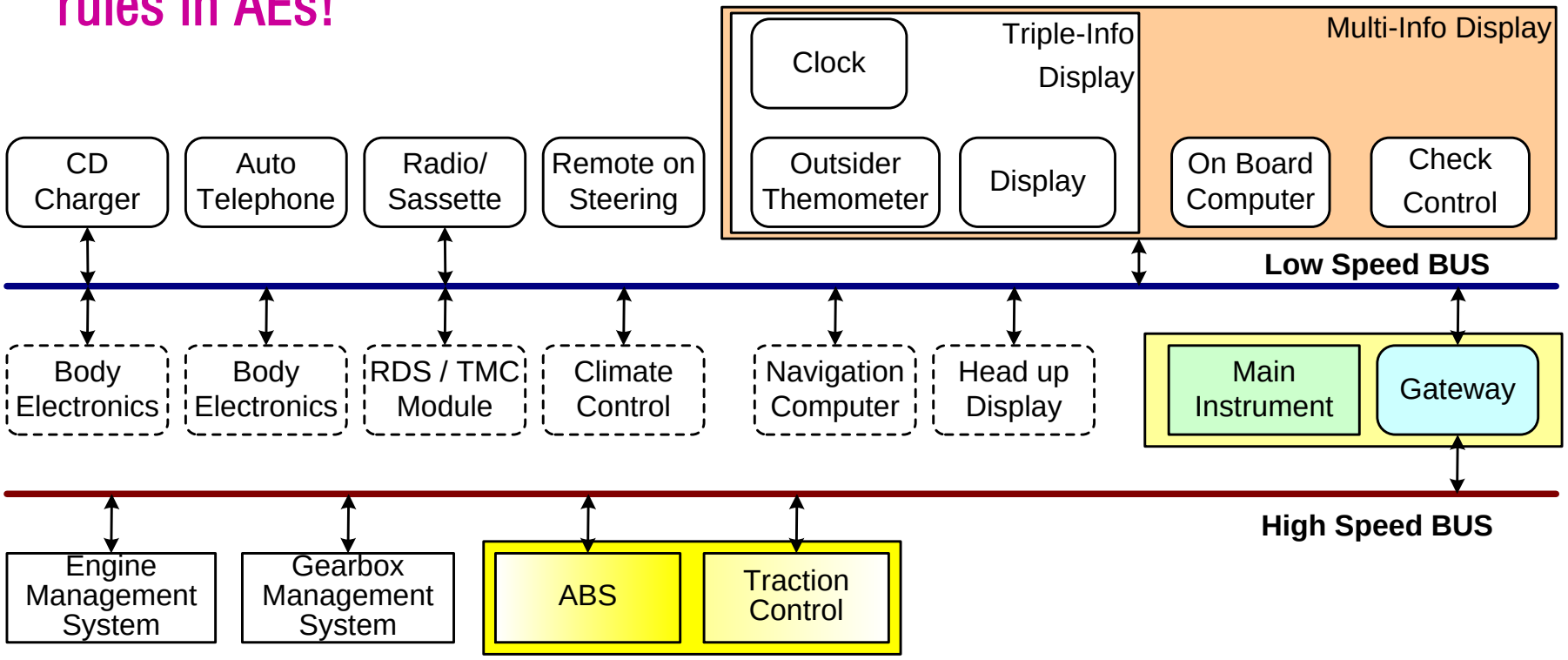




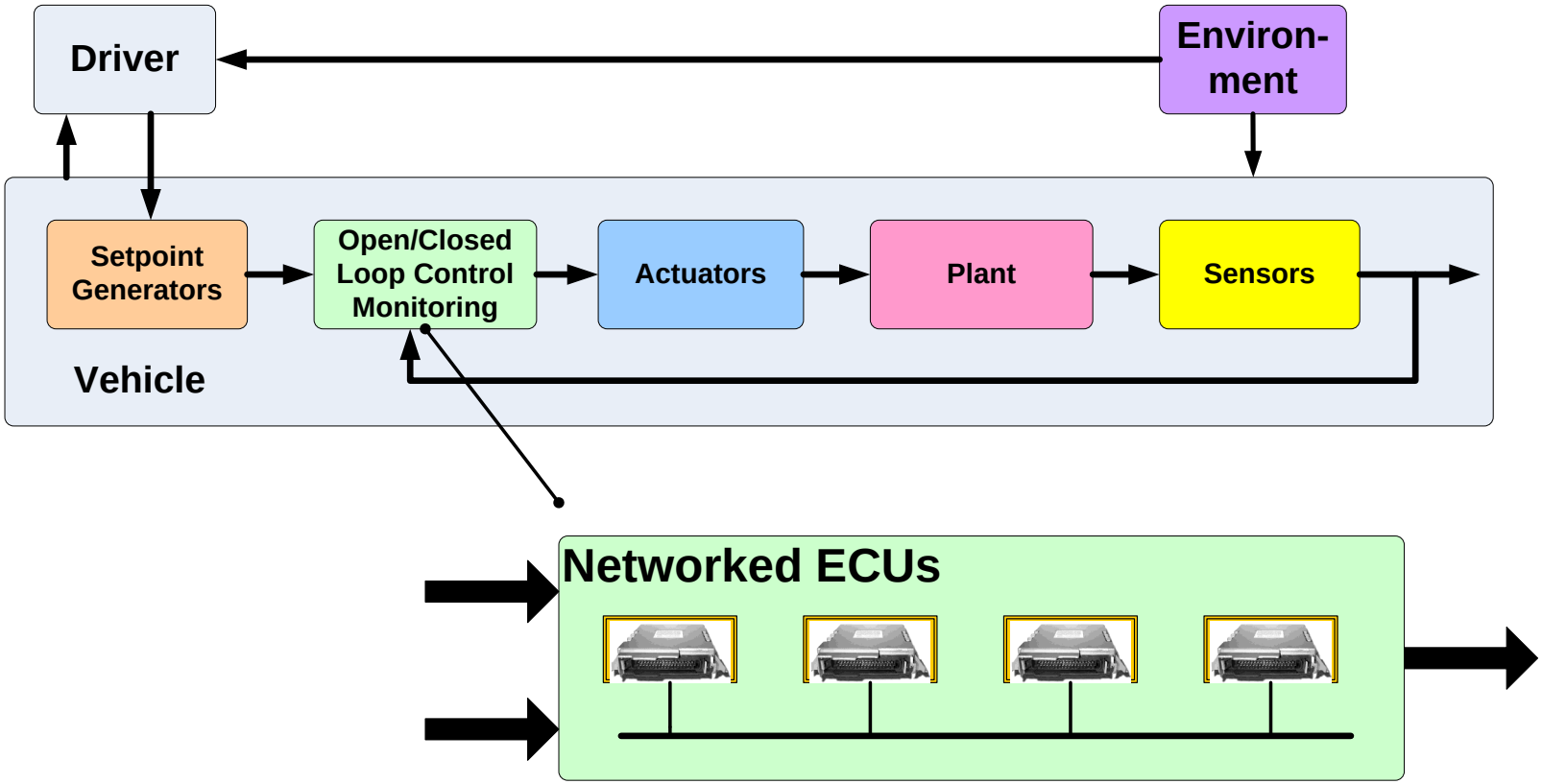
车辆电子控制系统

（2）汽车电子软件的核心开发流程 /Core Process of Software Development

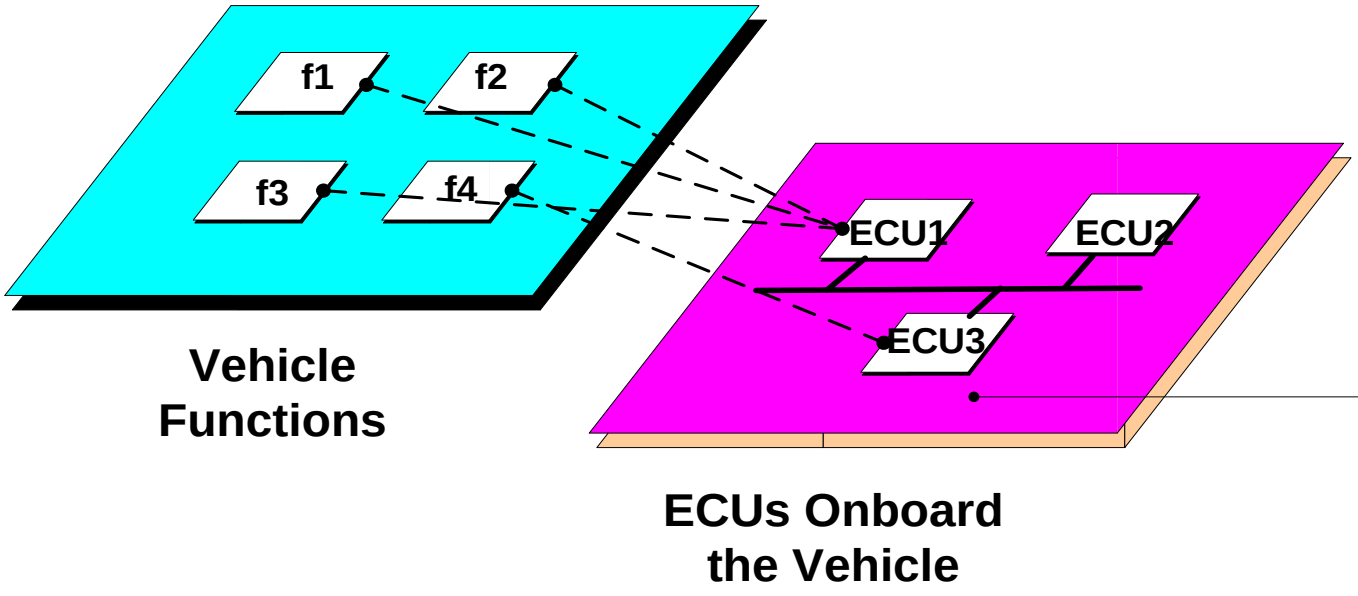
- 网络技术使得在控制系统中添加新功能更加容易，但也使得软件开发变得越来越重要/Network makes it easy to add new functions in control system. Software also becomes an essential rules in AEs!



（2）汽车电子软件的核心开发流程
/Core Process of Software Development



软件功能模块和网络化ECU
Software Function Mapping and ECU Network



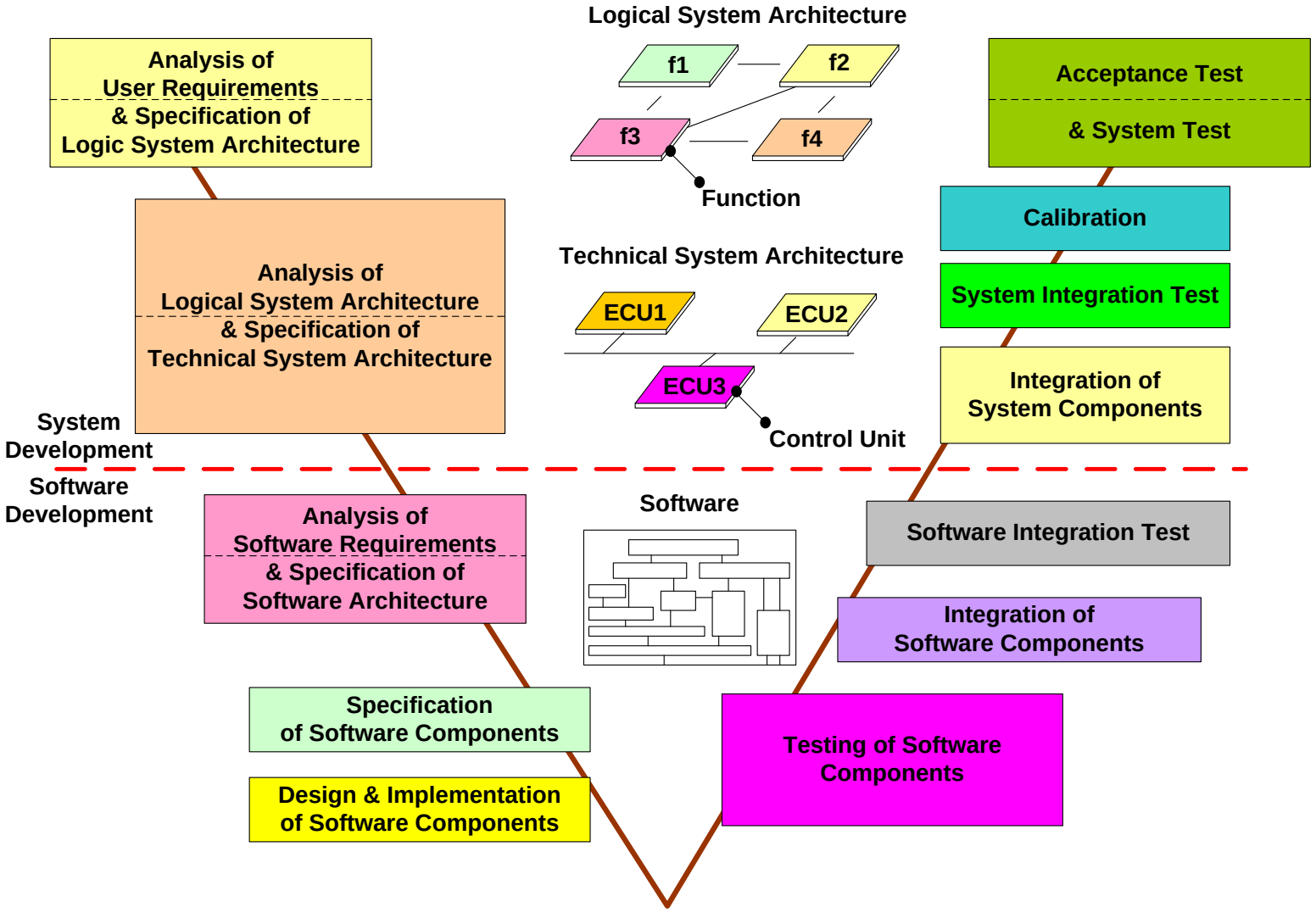


Vehicle

Function and ECU networks of the vehicle

（2）汽车电子软件的核心开发流程 /Core Process of Software Development

电控软件V型开发模式基本流程



（2）汽车电子软件的核心开发流程 /Core Process of Software Development

1. Analysis of user requirements and specification of logical system architecture; 在逻辑层次解决客户需求和指标
2. Analysis of the logical system architecture and specification of technical system architecture; 分析逻辑层次的构架，确定技术层次的解决方案；
3. Analysis of software requirements and specification of software architecture; 确定软件部分的解决方案和技术指标
4. Specification of software components; 确定软件模块的指标
5. Design, implementation, and tests of software components；设计和测试软件模块
6. Integration of software components and software integration tests；集成软件模块和软件集成测试
7. Integration of system components and system integration tests；系统集成和测试
8. Calibration；标定
9. System test and acceptance test；系统测试和检验

（2）汽车电子软件的核心开发流程 /Core Process of Software Development

- 质量控制的两个关键词：
  - Validation（确认）：是否达到了客户的要求?
    - Validation is defined as the process of evaluating a system or a component of the system to establish that it is satisfactory for its intended application and that it meets customer expectations. 可行性验证通过评估一个系统或者部件的过程，来判断其功能是否满足客户需求；
    - Function Validation as a process has the aim of establishing that the specification meets customer expectations and that it will have customer acceptance. 功能可行性论证是指建立满足客户期望，并被客户接受的功能定制过程。

（2）汽车电子软件的核心开发流程 /Core Process of Software Development

- 质量控制的两个关键词：
  - Verification（验证）：是否达到了技术指标要求?
    - Verification describes the process of evaluating a system or a component of a system to establish that the results of a given development phase meet the requirement for that phase. 验证的过程是指通过评估一个系统或者系统中一个部件，来证明某个开发阶段的结果是满足该阶段给定指标要求的。
    - Software Verification establishes that an implementation adequately meets the specifications defined for the respective development step. 软件验证要建立一套能充分证明软件满足该阶段的指标的实施过程。

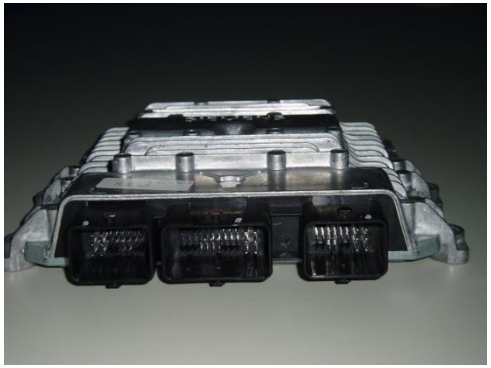
（3）功能集成/解耦和功能安全

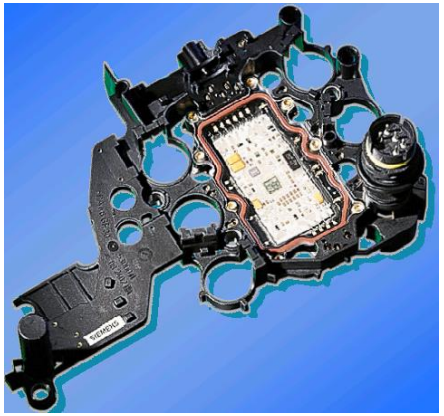
- **功能集成/ Functional Integration**

  将不同的功能有机地集成到一个部件中去/Implies the inter–linking of different sub–function into a single component.

  不仅只是将一个学科中的多个功能集成到一个部件，更要将机械和电气等多个学科的多个功能集成到一个部件中去。/The degree of the function integration is not limited only to the integration of several functions of one discipline, but rather serves to integrate many functions of mechanics and electronics into one interdisciplinary component.







（3）功能集成/解耦和功能安全

- 功能解耦/Function Separation:
  - 功能单元被分解为独立的子功能模块，这样便于优化/ Function units are separated in order to guarantee individual sub–functions.

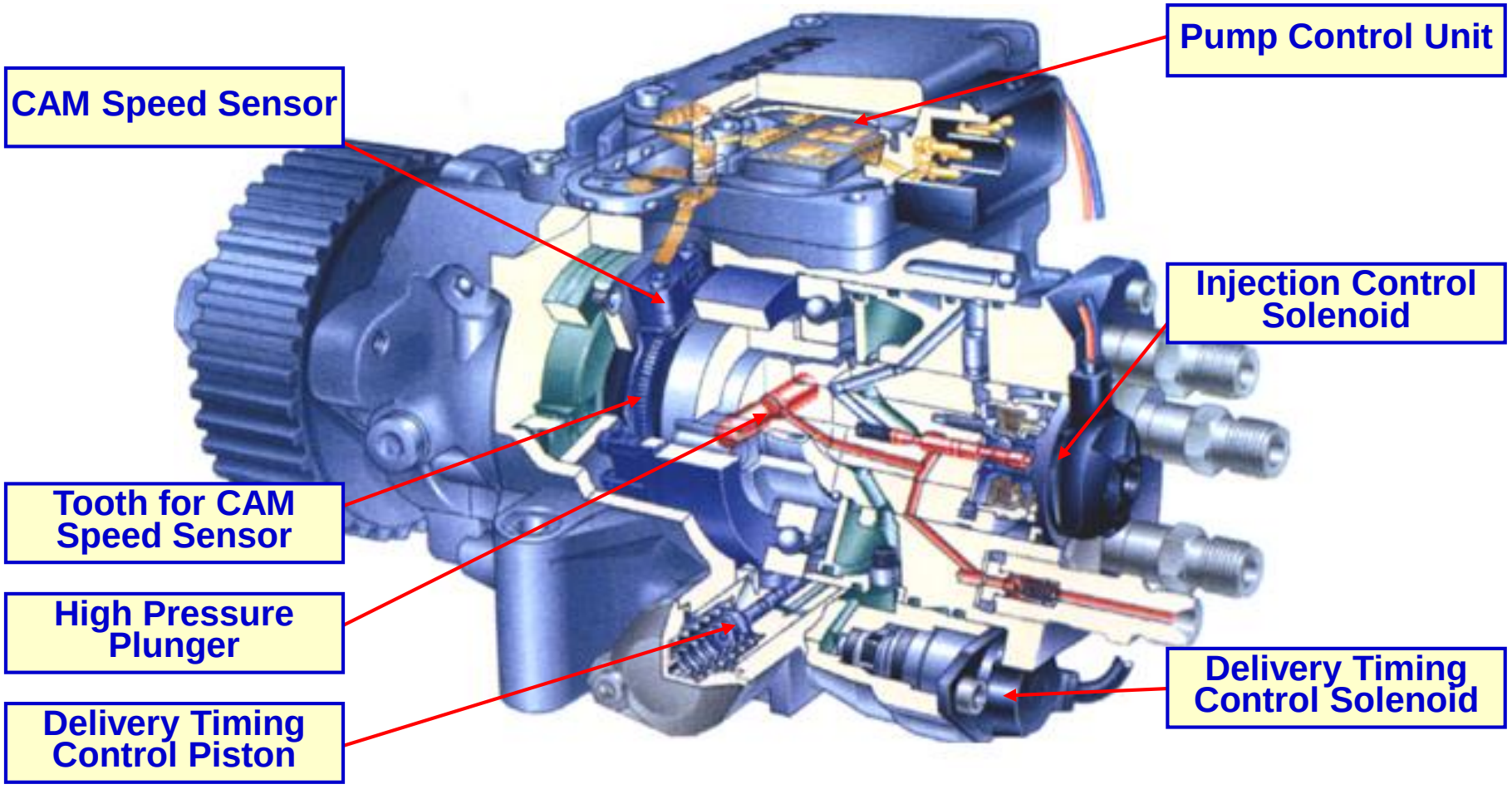
    目标是能够找到子功能的最佳解决方案，而不需要多约束折中/ One can use the best solution for the sub–function without make compromises.

    有的和安全相关的功能必须要单独考虑；在考虑电磁兼容性能时，功能解耦是非常重要的/ Some components relative to the safety can be considered individual. Function separation is also very important in the case of EMC.

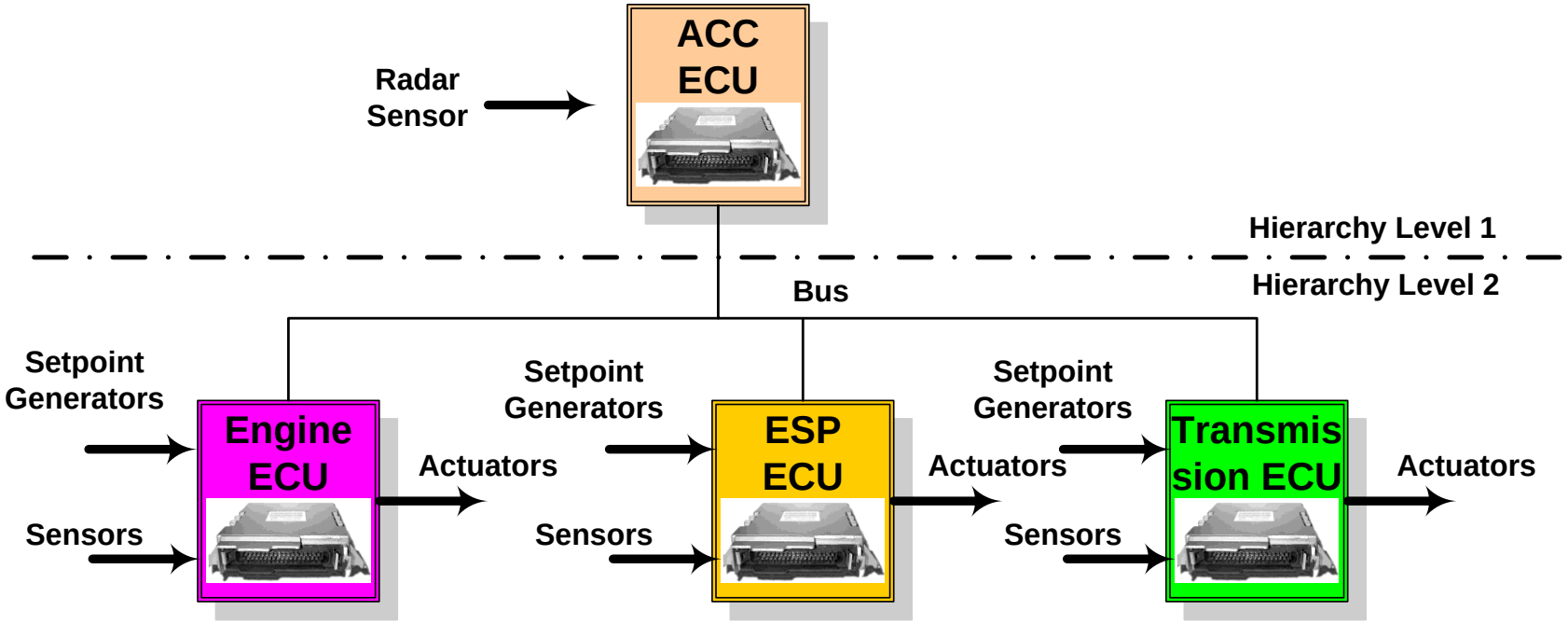
    功能解耦好处是，可以通过添加新功能的方法获得更多的扩展系统的性能/Finally, function separation is easy to add further optional functions according to the “ADD ON” approach.

    进一步，通过网络协议，实现了新功能添加和信息共享的最好方式/ Network is the best approach to achieve “ADD ON” Functions.

功能集成/Example of Function Integration

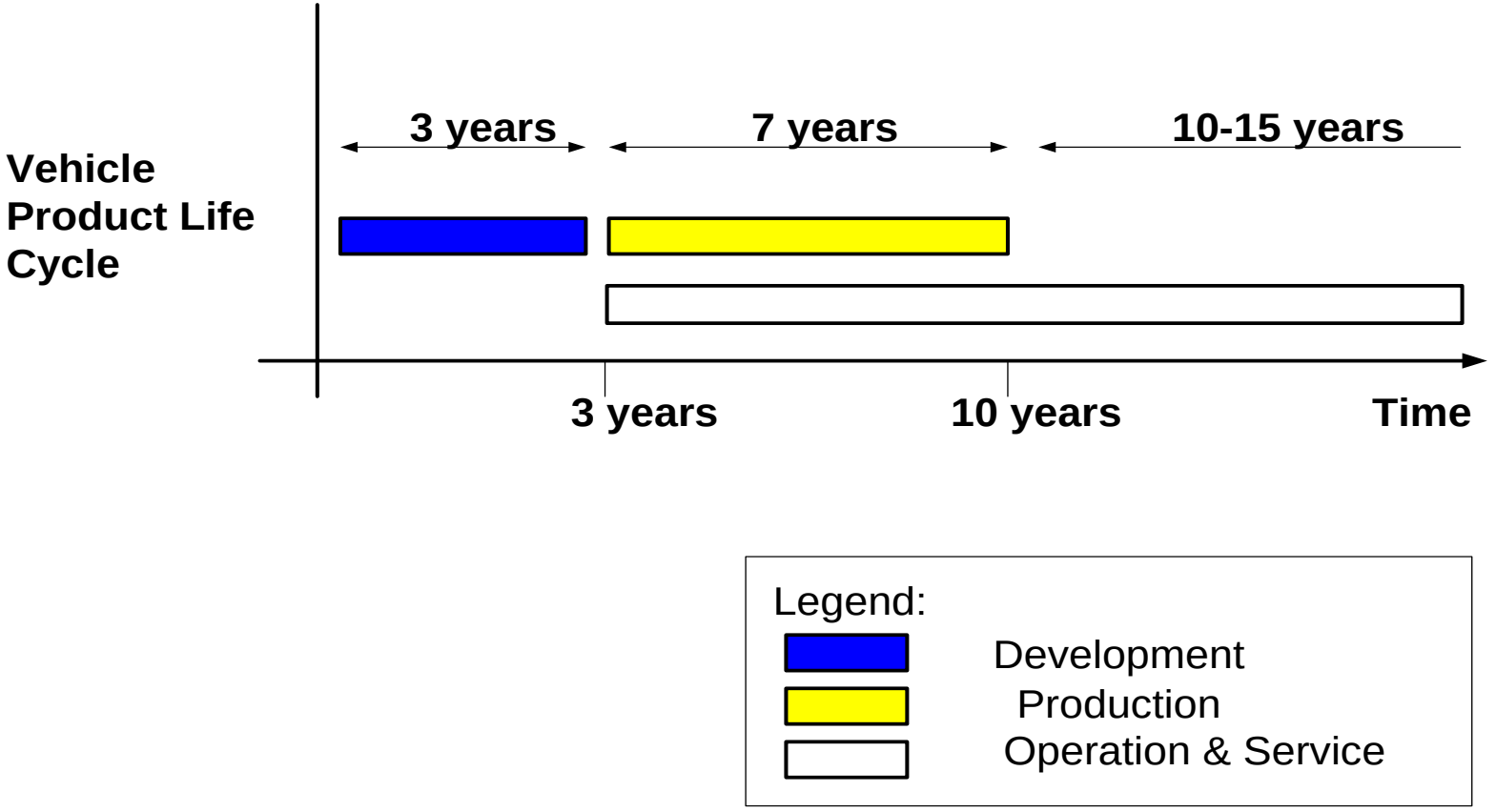


- 功能解耦：自适应巡航控制系统/Adaptive Cruise Control System



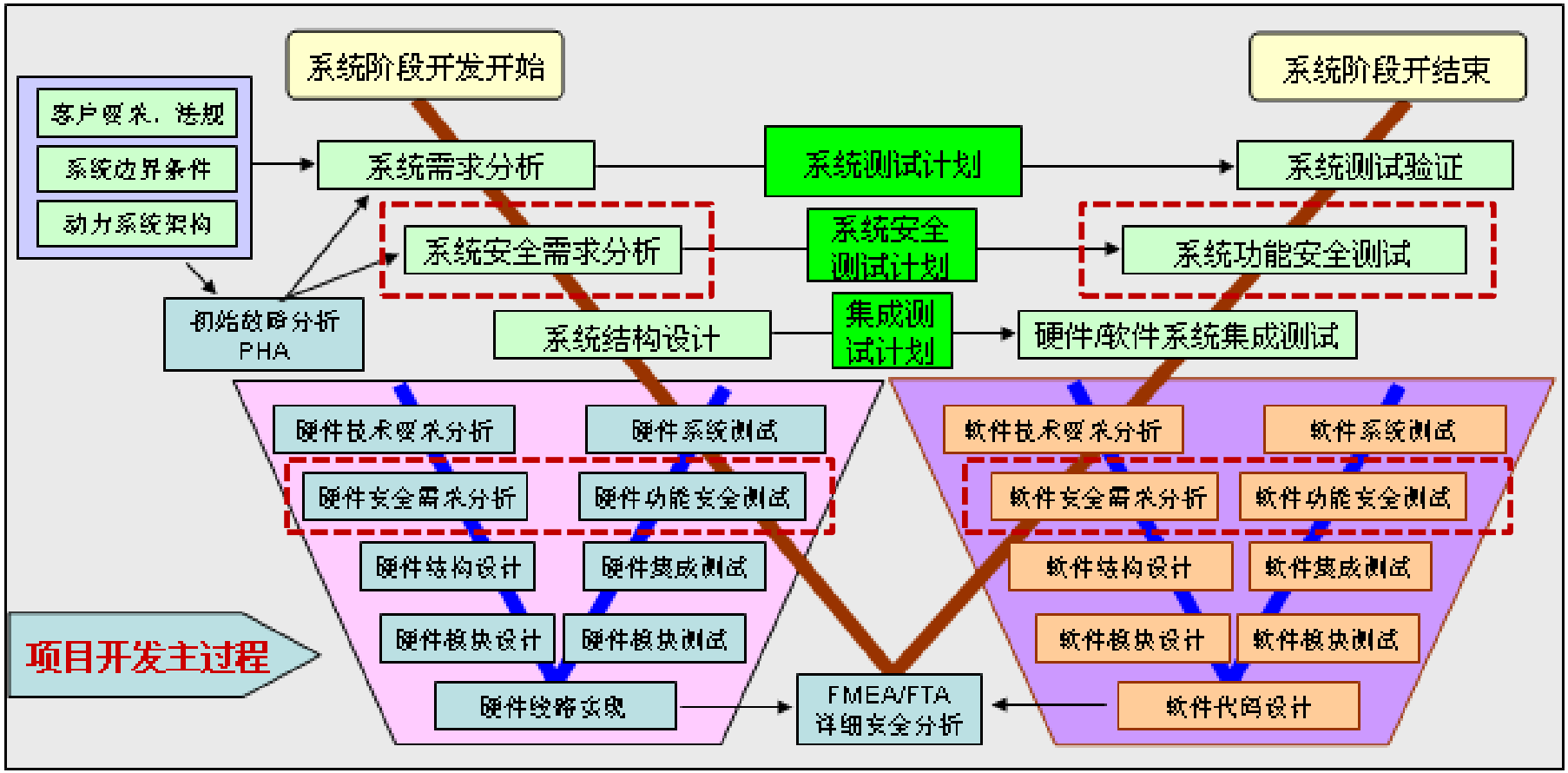
The ACC ECU and ACC Systems

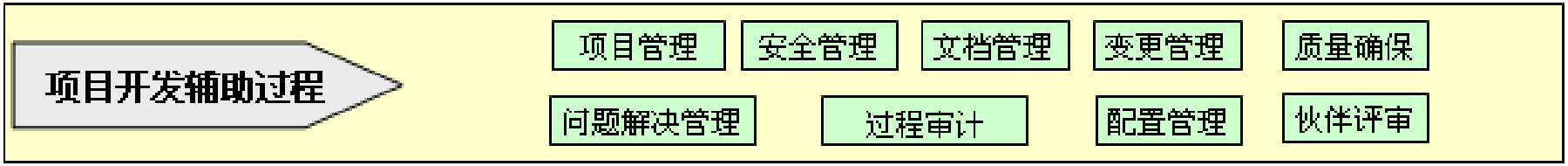
汽车电子产品的生命周期
/Production of Life Cycle of a Vehicle



（3）功能集成/解耦和功能安全

功能安全规范/Function Safety Standard：ISO26262





本节结束，谢谢！

Thank you！